



Venture capital exits in domestic and cross-border investments



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ABSTRACT

We compare and contrast the performance of cross-border and domestic venture capital (VC) investments in terms of the time it takes VC backers to exit portfolio companies through initial public offering (IPO), trade sale (M&A), or other routes. Initial analysis suggests that cross-border investments are exited more quickly than domestic investments, and this difference is driven by the cross-border investments in North America. We explore whether the difference between domestic and cross-border exits can be explained by regional variations in economic/market activity and legal systems. We find it is important to allow for the impact of these macro variables on the time to VC exit to vary across regions. This is crucial not just to capture the full impact of the macro factors, but also to explain the differences in time to exit between domestic and cross-border investments.

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1. Introduction

In this paper we study domestic and cross-border venture capital (VC) exits against the background of the growing internationalization of the VC industry and the substantial expansion of VC investments by domestic providers in foreign portfolio companies. Recent studies document the growth of cross-border VC flows. Aizenman and Kendall (2012) describe the continual rise in the internationalization of the VC industry, particularly over the past two decades. They document a rise in cross-border participation in VC and private equity deals from 15 percent in the early 1990s to over 40 percent in 2007. Schertler and Tykiová (2011) report that one third of worldwide VC deals are cross-border in the sense that they involve a VC firm or fund that is based outside the region of origin of the portfolio company.

Despite the growing importance of cross-border VC investments, there is little research to date on the performance of cross-border investments at the level of the individual VC investment or the individual portfolio company. This study aims to contribute to this limited body of literature by examining the

performance of cross-border VC investments as compared to domestic investments. Specifically, we focus on exit performance in terms of the time VC backers take to exit their investments (i.e., the VC investment duration). Our analysis explores whether cross-border exits are quicker or slower than exits from domestic VC investments after controlling for a range of known determinants of exit behaviour.

Gompers and Lerner (2001) point out that VC exit is an important aspect of the survival and growth of the VC industry. The exit prospect is critical to the VC firm: VC investment decisions are partly determined by the possibility of a timely exit because it allows VC firms to realize their returns (Pearce and Barnes, 2006; Cumming, 2008). Exit is an important source of funds for VC providers because it enables VC backers to re-cycle their funds into new investments, and (expected) successful exit determines the willingness of investors in VC funds to provide capital for the initial and follow-on funds (Black and Gilson, 1998). The time to exit is crucial. On the one hand, longer involvement with portfolio companies results in higher costs to VC backers of carrying (maintaining) the investment due to ongoing monitoring and illiquidity. On the other hand, longer investment duration allows VC backers more time to add value to the portfolio company. An understanding of the exit performance of cross-border investments relative to domestic investments is of great importance to VC firms and funds and to their investors in deciding whether and where to invest. It is

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also of interest to policy makers who may seek to influence cross-border VC flows. Note for instance the efforts by the European Union to encourage cross-border VC flows within Europe (The European Commission, 2013).

There are several studies that examine the time it takes VC firms to exit their portfolio companies. Giot and Schwienbacher (2007) provide evidence on the exit routes chosen by VC and private equity backers and the speed of exit from investments in domestic portfolio companies located in the US. Cumming and MacIntosh (2003), Cumming and Johan (2010) investigate the time to VC exit and its determinants in North America. Wang and Wang (2012) examine the impact of economic freedom in a country (controlling for a range of other factors) on the exit performance of foreign VC investors. Other studies examine exits by UK private equity backers (e.g., Jelic and Wright, 2011). To our knowledge, this paper is the first to compare and contrast exit (time) performance of cross-border VC investments with that of domestic investments.

In the US, most VC deals are domestically funded but in the UK and the rest of the world cross-border investments are prominent (Aizenman and Kendall, 2012). UK VC firms invest on average half of their funds in portfolio companies outside the UK (see Fig. 1 in the appendix). This study focuses on VC firms that are based in the UK and invest both domestically and abroad. We examine time to exit at the level of the individual VC investment (as opposed to the aggregate VC fund or country level), and this allows us to capture exit performance as experienced by individual VC firms and funds. We focus on exits by VC firms incorporated in a single country (UK) from portfolio companies in numerous countries, in order to investigate the comparative exit performance of domestic and cross-border investments and the impact on exit performance of regional effects and conditions in the country of origin of the portfolio company. VC investors may incur higher ongoing (marginal) transaction costs of maintaining cross-border investments as compared to domestic investments. All else equal, the higher marginal cost of holding cross-border investments may cause VC backers to exit cross-border investments more quickly than domestic investments.

For a sample of 4502 VC investments exited during 1990–2010, we study the time to exit through alternative routes (initial public offering – IPO, merger and acquisition – M&A, management buyout – MBO and liquidation).

In our initial multivariate analysis, we find that cross-border investments appear to be exited more quickly after controlling

for a range of known determinants of exit such as deal, company and VC firm characteristics. On closer inspection, we find that this result is largely driven by the significantly speedier exits in North America.

Previous studies suggest that the speed of exit may be linked to macro factors such as stock market and economic activity and the quality of the legal system (Black and Gilson, 1998; Cumming et al., 2006; Schwienbacher, 2008; Wang and Wang, 2012). Survey research by Schwienbacher (2008) finds that VC firms exit earlier from portfolio firms based in the US than from those based in Europe, and this may be due to more liquid markets in the US.

We explore the impact of macro factors controlling for the impact of North American investments, and find that market liquidity and legality speed up the time to IPO exit. M&A exits are speeded up by higher stock market valuations and gross domestic product (*GDP per capita*) slows down the time to exit for both M&A and IPO. Our analysis indicates the importance of controlling for the impact of North American observations to accurately assess the full impact of macro variables on exit times worldwide. For instance, failure to control for the differential impact of liquidity on time to exit in North America results in a significant underestimate of the accelerating effect of *Market Liquidity* on IPO exit across all the other regions outside North America. The dominance and distinctiveness of North American observations in cross-country samples is noted in other cross-country venture-capital studies (e.g., Cumming et al., 2010; Wang and Wang, 2012).

The rest of this paper is organized as follows: Section 2 discusses the research design including the research questions, data and methodology. Empirical findings are discussed in Section 3, while the conclusion is presented in Section 4.

2. Conceptual framework and research questions

There has been strong growth in the internationalization of the VC industry (particularly outside the US), partly due to large increases in the availability of VC funds worldwide and the greater attractiveness of foreign investments relative to those available domestically (e.g., Schertler and Tykvová, 2011). We find that UK VC firms invest just over half their funds in portfolio companies located outside the UK (see Fig. 1 in the appendix, and BVCA, 2010). In the light of the great and growing importance of cross-border investments, it is important

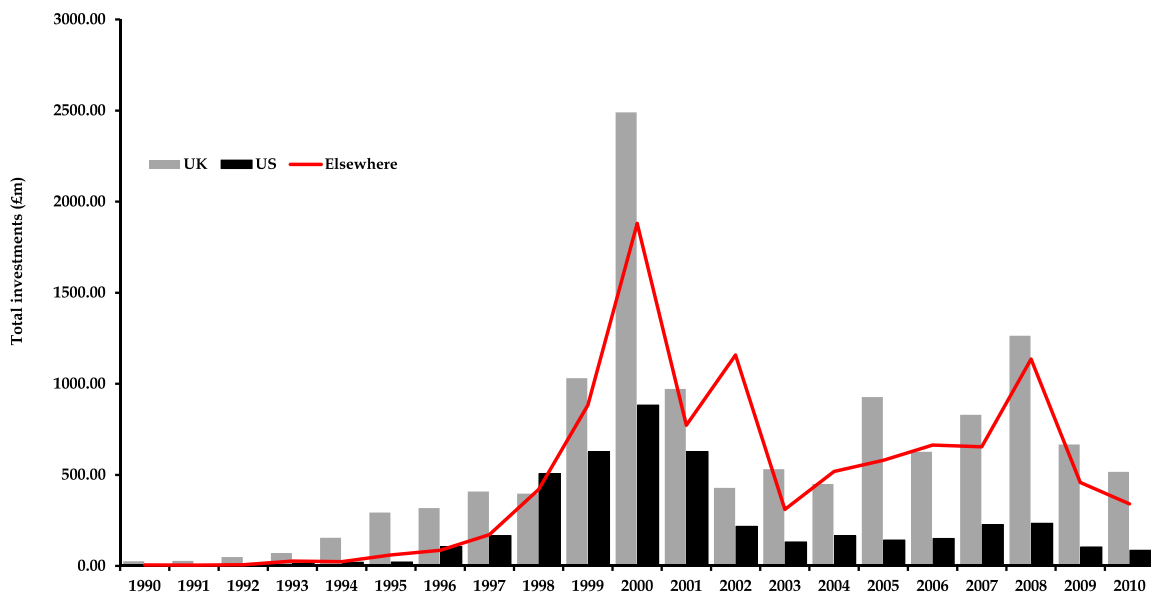


Fig. 1. Annual investments (£) by UK VC firms in the UK, US and the rest of the world during 1990–2010.

to understand whether there are differences in VC exit performance between domestic and cross-border investments, particularly in terms of the time to (successful) exit. This section discusses the role and determinants of VC exit, the potential sources of differences between domestic and cross-border investments in terms of exit behaviour, and alternative exit routes.

2.1. Explaining the time to VC exit

The cycle of exit and reinvestment is essential to the viability of the VC industry as exit allows VC firms to recover their funds and realize returns (Black and Gilson, 1998). The time to exit is an important measure of VC performance (Black and Gilson, 1998; Wang and Wang, 2012), and a key determinant of VC investment decisions and investors' willingness to provide capital for VC funds. VC firms may plan the (expected) time to exit by weighing up the costs and benefits of continuing with an investment. By providing not just financial but also reputational capital, as well as spending resources and effort on advising and monitoring portfolio companies, VC backers generate value and also incur costs. Both the (marginal) value added generated and the (marginal) costs incurred by VC backers are likely to vary systematically over the duration of the VC investment. VC backers may choose their optimal investment duration by trading off the marginal benefit (value added) of continued investment against the marginal cost.

The (marginal) value added by the VC provider continuing to back the venture providing advice, monitoring and reputation is likely to be high in the early stages of an investment and decline thereafter (Black and Gilson, 1998; Cumming and MacIntosh, 2003; Cumming and Johan, 2010). The VC backers' efforts, expertise and networks are likely to generate particularly high marginal value added in the initial phases of investment by guiding the most important strategic decisions of portfolio companies. As a result of their high initial input of (advisory and monitoring) effort, VC backers' expected marginal cost is also likely to be high initially but decline later (along with effort). VC backers will choose to exit as and when their marginal value added drops below their marginal cost.

2.2. Time to exit in cross-border versus domestic investments

While all VC investments require a high level of effort to be spent on advising and monitoring portfolio companies, greater distance between the VC backer and the portfolio company is likely to increase monitoring, information/communication and other transaction costs (Wang and Wang, 2012). Past findings suggest a VC investment bias in favour less distant and/or domestic investments (e.g., Cumming and Dai 2010; Aizenman and Kendall, 2012) and shorter times to (M&A) exit for less distant investments (Giot and Schwenbacher, 2007).

If VC firms determine their optimal time to exit by weighing up the costs and benefits of continuing an investment, greater monitoring and transaction costs of cross-border investments relative to domestic investments may tip the balance in favour of earlier exits from cross-border investments. Location bias may be reflected in a higher or steeper downward-sloping marginal cost curve in the framework outlined above. All else equal, this would lead VC backers to exit cross-border investments more quickly than domestic investments.

Domestic and cross-border investments may also differ in terms of the VC backer's expected marginal value added. Cross-border VC investors can help portfolio companies access technology and markets (product, financial, labour, etc.) in the VC investors' home country and possibly internationally (e.g., Mäkelä and Maula 2005, 2006; Devigne et al., 2013). If VC backing generates higher marginal value added in cross-border investments than in domestic investments for any given investment duration, all else equal we would expect cross-border investments to be exited later than

domestic investments (assuming marginal costs are the same for cross-border and domestic investments). On the other hand, cross-border VC backers may open up networks and opportunities that make the portfolio company more marketable. This may reduce the benefit (or increase the cost) of delaying investment and speed up successful exit through IPO or M&A.

In addition to IPO and M&A exits, we also examine less successful exits. Our reasoning on the possibly differences between cross-border and domestic investments in terms of exit times applies also to less successful exit such as (management) buyouts, including compulsory repurchases in the form of the buyback of the VC backer's shares by the entrepreneur, and liquidation. Existing evidence suggests that (unsuccessful) cross-border VC exit occurs more quickly as cross-border VC investors pull out more quickly than domestic VC backers in the light of disappointing portfolio company performance (Mäkelä and Maula, 2006).

In conclusion, there are reasons to expect cross-border investments to be exited either more quickly or more slowly (or at the same speed) as equivalent domestic investments depending on the relative marginal value added and marginal cost incurred by the VC backer. Since there are no reasons to favour either prediction, our (initial) analysis focuses on a test of the null hypothesis that there is no difference in the time to exit (investment duration) between cross-border and domestic VC investments.

2.3. The impact of markets and other macro factors

At exit, the VC backer expects to extract a large part of the value added to the portfolio company. The ability of the VC investor to extract this value added is likely to be affected by macro factors relating to market conditions, macroeconomic climate and the institutional framework of the portfolio company's home country. The macro factors may also affect the marginal (opportunity) cost to the VC backers from continuing an investment (Cumming and Johan, 2010).

Highly developed, liquid public capital markets allow VC backers to realise their investments and extract more of the value they added to the portfolio company. More liquid stock markets provide windows of opportunity for IPOs. As portfolio companies rush to go public during these windows, the time to IPO exit shortens. Stock market liquidity may also promote M&A exits by increasing the availability of external equity funding for acquisitions. Black and Gilson (1998) argue that more active stock markets (in the USA and other market-based economies) facilitate VC exit through an IPO. Differences across countries in terms of the level of domestic stock market activity have been shown to impact exit time and route. In a questionnaire study, Schwenbacher (2008) finds that domestic VC firms exit earlier in the US than in Europe and suggests that this may be due to stock markets being more liquid in the US. By contrast, Cumming et al. (2006) find no significant impact of country-specific stock market liquidity (measured by stock-market capitalisation) on IPO and M&A exits in 12 Asia-Pacific countries. Similarly, Wang and Wang (2012) report that liquidity (stock market valuation relative to GDP) is not significantly related to the likelihood of successful exit from cross-border VC investments worldwide. Based on these past findings, we tentatively expect higher stock market liquidity to speed up the time to exit, particularly for IPOs.

Relatively higher aggregate stock market valuations typically characterise windows of opportunity for IPOs and reduce the cost of external equity to potential acquirers. Higher stock market valuations have been shown to be associated with higher IPO and (public) M&A volumes (e.g., Shleifer and Vishny, 2003; Rhodes-Kropf et al., 2005). If companies and investors rush to exploit the window of opportunity generated by overvalued markets, we expect shorter times to exit. Buoyant stock market conditions also

increase the marginal opportunity cost to the VC backer of continuing a VC investment and hence reduce the expected time to exit (Cumming and Johan, 2010). The empirical evidence provided by Cumming and Johan (2010) from North American VC investments is consistent with the expectation that strong market conditions and high stock market valuations speed up VC exit.

We also expect the level of activity in the VC market to affect the time to VC exit. Greater VC activity in terms of higher demand for, and supply of, venture capital increases the opportunity cost to the VC backer of maintaining their funds in an existing investment as opposed to recycling them into promising new ventures. In such periods of high VC activity, VC funds and their investors (limited partners) are likely to push for speedier exit from existing investments (Black and Gilson, 1998; Cumming and Johan, 2010). The prediction that VC market activity speeds up exit is supported by the results of Cumming and Johan (2010).

The quality and effectiveness of legal systems are crucial to promoting the development of capital markets including IPO and VC markets. More efficient legal systems may mitigate agency problems between entrepreneurs and outside investors (Cumming et al., 2006). Better legal systems may also allow VC backers to extract more of their value added at the time of their exit. Better legal systems reduce the adverse effects of information asymmetries faced by new owners particularly in the case of IPO exits. Legality has been shown to positively affect the probability of an IPO exit for East Asian portfolio companies (Cumming et al., 2006). Over time, a higher exit probability in each period should translate into a shorter time to exit. Hence, we expect that better legal systems reduce the time to exit through an IPO.

Differences in macroeconomic conditions are likely to impact VC exits. Higher levels of economic activity or development are associated with IPO and M&A waves (e.g., Dittmar and Dittmar, 2008). Cumming et al. (2006), Cumming and Johan (2008) and Cumming et al. (2010) find that high levels of economic activity and output lead to higher probabilities of exit, particularly through the IPO and M&A routes. Over time, higher exit probabilities should translate into shorter times to exit. On the other hand, the results of Wang and Wang (2012) suggest that country-specific economic development (measured as GDP per capita) delays VC exit. They interpret the observation that poorer countries outperform richer ones in terms of achieving faster VC exits as consistent with the 'convergence hypothesis' which predicts that poorer countries eventually catch up with richer countries. Due to the conflicting existing evidence, we leave the direction of the relationship between GDP per capita and the time to exit to be determined empirically.

2.4. Impact of VC investor and portfolio company characteristics

VC firms differ in terms of their abilities, expertise and specialization (linked, e.g., to economies of scale) and the characteristics of VC firms are likely to affect their marginal value added and marginal cost. Some VC firms may have stronger incentives and capabilities of achieving speedy exits than others. For instance, less seasoned VC firms may have higher marginal (opportunity) costs of continuing investments. Young VC firms have been shown to push for quick exits in an effort to 'grandstand' and attract potential future investors in follow-on funds (Gompers, 1996). Conversely, more experienced VC firms may have the expertise and networks to facilitate speedier exits. Similarly, funds that are near the end of their lifespan (when they will need to return money to their limited partners) may be forced to exit quickly, potentially even before their marginal value added drops below their marginal cost. Following the standard approach in the literature (e.g. Cumming et al., 2006; Wang and Wang, 2012), our analysis controls for the characteristics of VC firms and VC funds in order to differentiate the impact of these characteristics from the effects of the cross-border or regional locations and of the macro variables.

We also expect differences among portfolio companies to affect the marginal value added and cost of VC backers, and hence the expected time to exit. Some portfolio companies, specifically larger, more mature companies, or companies in certain industries, are inherently more capable of achieving quick successful exits. Portfolio companies attracting cross-border funding are likely to have greater international visibility possibly due to their size, age, industry, etc. To account for the resulting differences between the domestic and cross-border subsamples, we control for portfolio company characteristics and industry classification. The control variables in terms of the VC firm, fund and the portfolio-company characteristics are discussed in further detail in Section 3. Table 1 summarises the definitions of all variables.

2.5. Exit routes

Among the alternative exit routes available to VC firms, exit through an initial public offering (IPO) is typically considered the most successful type of exit (Black and Gilson, 1998) followed by exit through a merger or acquisition (M&A) such as a trade sale. Previous studies including Cumming (2008), Espenlaub et al. (2014) find that VC backers earn higher returns from IPO exits than from M&A exits. Returns to VC firms are typically lower when an exit occurs through a (management) buyout, possibly in the form of a compulsory buyback of VC stakes by the entrepreneur. Liquidations typically result in the loss of all of the VC investment, i.e., a return of –100% (Cumming, 2008). Our analysis considers all four exit routes, with a special focus on the two successful routes, M&A and IPO.

3. Methodology and data

3.1. Econometric methodology

Modelling VC time to exit through different routes requires a dynamic framework such as that provided by survival (duration) models. These models measure the time that elapses between two events. In our analysis, this involves measuring the time from the date on which a VC firm invests in a given portfolio company to the date on which the VC firm exits this investment. The estimation technique corrects for censored observations; in our study censoring occurs when an investment has not been exited by the end of the study period. Duration analysis incorporates both censored and uncensored observations to provide consistent estimators. Shumway (2001) finds that survival models outperform static models (i.e., logit and probit models) in terms of out-of-sample forecasts. Survival analysis has been applied to financial data in various ways, including the modelling of stock delistings (Esenlaub et al., 2012; Hensler et al., 1997; Jain and Kini, 2000) and VC exits (Giot and Schwenbacher, 2007).

We estimate cumulative VC exit rates non-parametrically using the Kaplan–Meier method (e.g., Espenlaub et al., 2012). The exit function, normally termed the failure function, $F(t)$, is defined as the complement of the survival (i.e., continued investment) function $S(t)$:

$$F(t) = \Pr(T \leq t) = 1 - S(t) \quad (1)$$

$$S(t_j) = \left(\frac{n_j - d_j}{n_j} \right) S(t_{j-1}) \quad (2)$$

Eq. (2) expresses the survival function $S(t_j)$ in month t_j as the probability of the VC investment continuing (i.e., surviving) beyond t_j conditional on the investment not having been exited until at least month t_j , multiplied by the survival function in the previous month t_{j-1} (see Kleinbaum, 1996, p. 56); n_j is the number

Table 1
Definitions of variables.

Variable	Definition of variable and unit of measurement
VC Age	VC age measured as (the logarithm of) the difference (in years) between the founding date and the date of investment. In the case of multiple VC providers, we measure the average of their ages.
Old VC	VC age of (the logarithm of) the oldest 25% of VC firms, measured as VC Age interacted with a binary dummy variable coded 1 for observations with (average) VC Age above the 75th percentile, and 0 otherwise.
VC Size	Size of the VC firm measured by (the logarithm of) the amount (in millions of 2006 GBP) invested by the VC firm in all its portfolio companies in the calendar year before the year of exit. In the event of no exit, the variable is measured at the end of 2010. In the case of multiple VC providers, we measure the average size.
Fund Age	Fund age measured as (the logarithm of) the difference in years between the VC funds first investment (in any company) and the date of exit from the portfolio company for which we observe an exit. In the event of no exit, the variable is measured to the end of 2010. In the case of multiple VC funds investing in the portfolio company, we measure the average age of the funds.
Old Fund	Fund age of the oldest 25% of funds, measured as (the logarithm of average) fund age interacted with a binary dummy variable coded 1 for observations with fund age above the 75th percentile, and 0 otherwise.
Size Inv	The (logarithm of the) amount invested in millions of 2006 GBP by the VC provider in the given portfolio company for which we observe an exit. In the case of multiple providers, the variable is defined as the average investment across the multiple VC providers.
TOP25% Size Inv	The highest 25% of the (average) amount the VC invested in a portfolio company, i.e., Size Inv interacted with a binary dummy variable coded 1 for observations with Size Inv above the 75th percentile, and 0 otherwise.
Synd	The number of VC funds syndicated in the first financing round of the portfolio company for which we observe time to exit.
Old Synd	Indicator of the presence of an old VC fund in the syndicate. Measured as Synd interacted with a binary dummy variable coded 1 for observations with fund age above the 75th percentile, and 0 otherwise.
Company Age	The (logarithm of the) age of the portfolio company, measured at the time of VC investment as the difference (in years) between the founding date and the investment date.
Company Size	The (logarithm of the) book value of assets of the portfolio company at the time of VC investment (in millions of 2006 GBP).
MTB	The market-to-book ratio (measured as the market value divided by the book value of equity) of the stock exchange(s) in the country of origin of the portfolio company in the calendar year before the year of the exit (Source: MSCI).
Market Liquidity	The market liquidity in the country of origin of the portfolio company measured in the calendar year before the year of the exit as the ratio of total market capitalization of equity traded on the company's stock exchange divided by its gross domestic product (Source: World Bank).
Deal Activity	The (logarithm of the) number of VC deals in the country of origin of the portfolio company in the calendar year before the year of the exit. In the event of no exit, the variable is measured for the year of investment.
Legality	The index of the quality of the legal system in the country of origin of the portfolio company (Berkowitz et al., 2003) in the calendar year before the year of the exit; the index is constructed as the weighted average of separate indicators, used in La Porta et al. (1998), of the efficiency of the company's judicial system, the rule of law, corruption, the risk of expropriation, and the risk of contract repudiation.
Country	For a given portfolio company, binary dummy variables indicating the (region of the) country of origin of the portfolio companies. We consider the following five regions and use four dummy variables (with UK & Ireland as the base): • US & Canada (North America) • UK & Ireland • Continental Europe • Latin America • Asia & Africa

Please note that the variable labels defined above are used as de-logged variables in the descriptive statistics but in the estimation analysis the variables are used in their logarithmic form (as natural logarithms).

of investments that have not been exited by the start of month t_j (also known as the risk set at t_j), and d_j is the number of investments exited during month t_j .

Below, we apply Eqs. (1) and (2) to estimate cumulative exit rates for VC investments. We calculate exit rates by geographic region. We also estimate exit rates separately for four mutually exclusive exit routes: trade sales (M&As), IPOs, MBOs (including compulsory buybacks) and liquidations. To test whether investments in different groups (such as subsamples grouped by region or exit method) share the same Kaplan–Meier survival curves, we use the log rank test (e.g., Kleinbaum, 1996, pp. 557–563), a large-sample chi-square test. The test involves comparing observed and expected exit rates. If the observed exit rate is significantly different from the expected rate, the test rejects the null hypothesis that the groups share the same survival curves.

To examine factors that accelerate or delay the exit times, we use an Accelerated Failure Time (AFT) frailty model. The model corrects for ‘frailty’ in the sense of heterogeneity (mixed distributions) due to unobservable variation across VC firms in terms of their skills, expertise and networks, similar to the way in which a regression model corrects for heterogeneity or fixed effects. We use clustered standard errors to control for correlation among the exits of a given VC firm. The frailty model can be thought of as an AFT survival model with an additional parameter (theta).² There are

various types of frailty model, and a log likelihood ratio can be used to select the best-fitting model for a given set of data. In our analysis, we use the Weibull-gamma distribution, which we find to have a larger log likelihood ratio than the exponential-gamma and lognormal-gamma distributions.

To examine the determinants of a specific exit route relative to another, we also estimate a competing risk model (as proposed by Fine and Gray, 1999). In a competing risk model it is possible to measure the effects of determinants on, say, IPO exits relative to, say, M&A exits by defining the latter (M&A) route as the competing event. Exits through routes that are not modelled (e.g., in the IPO/M&A model, buyouts and liquidations are not modelled) are treated as censored observations.

3.2. Cross-border, regional and macro factors

We aggregate the portfolio companies’ countries of origin into five regions: UK & Ireland, US & Canada (North America) Continental Europe, Latin America and Asia & Africa. Using UK & Ireland as the base, we include in our analysis regional (intercept) dummy variables for the latter four cross-border regions.

To capture the impact of market, economic and legal conditions on the time to exit, we use three measures of market liquidity, activity and valuation. First, we include market liquidity computed as the total market capitalization of the shares traded on a country's stock market(s) divided by gross domestic product (Demirguc-Kunt and Levine, 1996; Wang and Wang, 2012).

² Using a likelihood ratio test, we reject the null hypothesis that theta is zero; hence, we use a frailty model.

Second, we measure deal activity as the number of VC deals in the region. Third, we include the market-to-book ratio of the regional stock market(s). Following Cumming et al. (2006) we include the legality index developed by Berkowitz et al. (2003) based on the legal and financial variables of La Porta et al. (1998). This index is a weighted average of indicators of the efficiency of the judicial system, the rule of law, corruption, risk of expropriation and contract repudiation, and shareholder rights. A higher legality index reflects a better legal system. We measure the macro variables in the year prior to the year of exit and aggregate them into regional measures by averaging across the countries in a given region (for a given pre-exit year).³ As a result, the market, economic and legal measures vary across regions and over time.

Our analysis controls for other variables that influence VC exit, specifically the characteristics of the VC provider and the portfolio company. These control variables are discussed in detail below.

3.3. VC characteristics

As noted in Section 2, the characteristics of VC firms (their abilities, expertise and specializations) are likely to affect their marginal value added and marginal cost and hence their exit decisions. Some VC firms may have stronger incentives and capabilities of achieving speedy exits than others. For instance, young VC firms that lack proven track records of investment success may try to exit their investments speedily to raise their profiles ('grandstand') among prospective investors in their follow-on funds (Gompers, 1996). It is reasonable to expect that grandstanding is most likely to involve exit routes with high public visibility and impact, such as IPOs. Therefore, we might expect younger VC firms to exit more quickly through an IPO. By contrast, Giot and Schwienbacher (2007) argue that more experienced VC backers help facilitate and speed up successful exits through their greater expertise and wider networks. In light of these arguments it is not clear whether VC experience as measured by VC age will accelerate or delay the time to IPO or M&A exit. Similarly, VC experience (age) may accelerate or delay liquidation. On the one hand, more experienced VC firms may have superior skills in adding value to unpromising portfolio companies. On the other, they may spot underperforming portfolio companies more quickly and push for speedy liquidation rather than risking their reputation by continuing to back unpromising ventures. Similar reasonings apply to the size of the VC firm, which is another measure of its experience, reputation and the size of its network of contacts.⁴ We define VC size in terms of the total amount of capital committed by a VC firm to all the companies in its portfolio.

We expect VC fund age to have a significant, accelerating impact on the time to exit. VC funds have a predetermined, limited lifespan of 10 years (Pearce and Barnes, 2006; Cumming and Johan, 2008) with possible extensions of between one and three years. Hence, as a VC fund approaches the end of its life, the need to exit from its investments becomes ever more urgent. The impact of fund age on time to exit may not be linear and we expect that the oldest funds should have a particularly high impact on time to exit. Old funds will have to liquidate their investments at fund maturity (Kandel et al., 2011) even if the marginal value added by the VC backer continues to exceed its marginal cost. We

therefore expect the oldest funds to have a particularly pronounced, accelerating impact on exit.

3.4. Deal and portfolio company characteristics

In addition to VC firm and fund characteristics, we also expect deal characteristics, syndication and deal size, to impact the time to exit. Cumming and Johan (2010) argue that syndication may reduce the expected investment duration due to conflicts of interest among syndicate members that raise the marginal cost of continuing the investment. In the absence of such conflicts, larger syndicates allow greater sharing of investment risk and of the costs of monitoring and advising the portfolio company. This lowers the individual backer's marginal cost and may delay exit. On the other hand, free-riding among larger syndicates may reduce the marginal value added generated by individual members and thus shorten expected time to exit. Previous studies suggest that larger syndicates facilitate successful trade sale and IPO exits by providing valuable contacts, connections, and quality certification (Megginson and Weiss, 1991; Lerner, 1995; Brander et al., 2002; Giot and Schwienbacher, 2007). Hence, we expect a larger VC syndicate size to reduce the time to exit through an M&A or IPO.

A larger syndicate may also facilitate exit through liquidation or buyout (MBO) by exerting greater pressure on the entrepreneur(s) to liquidate or buy back unsuccessful ventures. Exit through liquidation may also be accelerated if unproductive conflicts among members in larger syndicates cause the portfolio company to fail more quickly.

The size of the VC investment in a portfolio company is expected to influence the decision to exit the portfolio company. VC firms tie up substantial amounts of capital in illiquid investments in portfolio companies. Larger investments may generate higher marginal costs for VC backers of continuing to hold the illiquid investment, and may be expected to be exited more quickly (Cumming and Johan, 2010). A large investment may also reflect the VC firm's confidence in the future success of a portfolio company, and their expectations of quick, successful exit from such large investments. Giot and Schwienbacher (2007) find that the larger the amount of money committed by VC backers to a (US) portfolio company, the speedier is the VC exit. Cumming and Johan (2010) also report shorter times to exit for larger US deals. As a result, we expect larger investment size to be associated with shorter times to successful exit.

VC firms typically invest in relatively young, small companies. Older and larger the portfolio company are likely to provide less scope for VC backers to add value as the companies have relatively well-developed networks, reputational capital, governance and organisational structures (Cumming and Johan, 2010). Larger and older companies also tend to have lower information asymmetries between insiders and outside investors which facilitate successful exit (Cumming et al., 2006). Therefore, we expect speedier exit through M&A or IPO when the portfolio company is larger or more mature. By contrast, smaller and younger companies involve greater information asymmetries and risk, and may be more prone to (and exited more quickly through) liquidation.

The marginal value added and cost of VC backers, and hence the expected time to exit, may vary systematically across industries (in terms of the industrial classification of the portfolio company). Certain industries, such as high tech, tend to provide VC backers greater scope to add value to portfolio companies than more traditional industries, and expected times to exit are longer in these industries (Cumming and Johan, 2010). As the scope for VC backers to add value is also higher in earlier stages of financing, we expect longer times to exit in early-stage investments as opposed to expansion or later stage deals.

³ The objective of this study is to empirically capture the impact on the time to exit of market, economic and legal conditions around the time of exit. One limitation of this approach (i.e., of using measures observed near exit rather than at the time of VC investment) is that our model cannot be used to predict the time to exit at the time of investment.

⁴ As shown in Table A1 in the Appendix, VC size and age are significantly positively correlated.

Finally, the exit decisions of VC firms are likely to vary over time. As a result, our analysis controls for year (of exit) effects.

3.5. Data

We observe all exited and un-exited investments by UK-based VC backers during 1990 through 2010 available from VentureXpert. Syndicated investments are included. We impose two sampling criteria. First, to be included in the sample we require the following data to be available: exit date, size of investment, company value post-financing, funds first investment date, and the founding dates for the portfolio company and the investing VC firm(s). Second, for companies with multiple financing rounds, we exclude all but the first financing round to avoid giving undue weight to these sample companies. The final sample consists of 4502 VC investments with a complete set of data; these investments were made by 477 UK VC firms. If VC firms exit only partially at the IPO, we classify this as an IPO exit and treat the IPO date as the exit date. We track each investment until VC exit or December 2010, whichever occurs earlier.

Following VentureXpert, we categorize our observations (of the first VC financing round in a given portfolio company) into one of the following six stages of financing: start-up/seed, early stage, expansion stage, later stage, buyout, and other financing stages. We define venture capital to comprise the first four financing stages (start-up/seed, early stage, expansion stage, later stage). We exclude from our sample financing rounds classified by VentureXpert as buyouts because our focus is on VC deals that are typically much smaller and younger investments than buyout deals.⁵ We also exclude 217 other rounds. According to VentureXpert, information about these rounds is not disclosed by the VC firms so that we cannot be sure whether they are VC or buyout investments. We aggregate start-up/seed and early stage into a single stage, and include three dummy variables for VC financing stage in our analysis.

Table 1 details the variable definitions, data sources and units of measurement. *VC Age* is measured as the (natural logarithm of the) difference between the founding date of the VC firm and the date of investment. *Old VC* is equal to *VC Age* for the oldest 25 percent of VC firms, i.e., for observations in the top quartile of *VC Age*, and zero otherwise. The age of the VC fund (*Fund Age*) is measured as the (logarithm of the) difference between the funds first investment date and the date of exit. *Old Fund* equals *Fund Age* for the oldest 25 percent of funds (in the top quartile of *Fund Age*), and zero otherwise. The size of the VC firm (*VC Size*) is measured by the (logarithm of the) amount invested by the VC firm in all its portfolio companies in the calendar year before the year of exit from a given portfolio company. The average size of the VC investment, *Size Inv*, is the (logarithm of the) amount invested by a VC backer in the portfolio firm. *TOP25% Size Inv* equals *Size Inv* for the top 25 percent in terms of the average size of VC investment (those in the top quartile of *Size Inv*), and zero otherwise. For syndicated deals, the VC characteristics are measured as the averages of the syndicated VC firms. The size of the VC syndicate (*Synd*) is the number of VC firms syndicated in the first financing round of a portfolio company. *Old Synd* equals *Synd* if the syndicate of funds investing in a portfolio company includes any of the oldest 25 percent of funds (i.e., those in the top quartile of *Fund Age*), and zero otherwise.

The portfolio company age (*Company Age*) is measured as the logarithm of the time (in years) from the company founding date to the date of the VC investment. The size of the portfolio company

Table 2

Variables relating to countries of origin of portfolio companies.

Country	No of obs	MTB	Market liquidity	Deal activity	Legality
Argentina	8	1.069	0.388	5	12.34
Australia	9	2.057	0.641	98	20.44
Austria	15	2.162	1.044	20	20.76
Belgium	17	2.051	1.012	38	20.82
Brazil	18	1.834	0.556	27	14.09
Canada	216	2.293	1.255	631	21.13
Chile	11	1.124	0.415	2	14.70
Denmark	19	2.285	1.236	47	21.55
Egypt	10	1.092	0.357	2	11.34
Finland	3	2.012	1.658	80	21.49
France	176	2.164	2.118	250	19.67
Germany	187	2.258	2.802	165	20.44
Greece	2	1.787	0.954	3	14.91
Hong Kong	19	2.051	1.223	20	19.11
India	77	2.083	1.265	103	12.80
Indonesia	8	1.476	0.948	3	9.16
Ireland	63	1.521	0.948	38	18.92
Israel	43	1.846	1.217	62	16.54
Italy	31	2.095	2.015	32	17.23
Japan	12	2.001	1.825	52	20.36
Kenya	11	1.517	0.314	2	12.00
Malaysia	14	1.806	0.879	8	16.67
Mexico	8	1.454	0.410	4	12.82
Netherlands	35	2.384	2.255	65	21.67
Nigeria	9	1.645	0.394	4	9.39
Norway	16	2.251	1.020	30	21.78
Philippines	7	1.121	0.345	3	8.51
Portugal	2	2.015	0.673	26	17.20
Singapore	24	2.042	0.703	24	19.53
South Africa	15	1.837	0.548	8	14.51
South Korea	1	2.045	0.675	210	14.23
Spain	41	2.095	1.023	54	17.13
Sri Lanka	12	1.028	0.325	1	10.40
Sweden	75	2.107	1.851	101	21.56
Switzerland	27	2.310	1.394	32	21.91
Taiwan	5	1.425	0.238	22	17.62
Turkey	18	1.689	0.644	2	11.84
UK	2049	3.185	2.797	348	20.41
US	1088	3.590	3.821	4003	20.85

For each of the countries of origin of the portfolio companies in our sample, this table shows market-to-book (MTB) ratio of the company's stock exchange, stock market liquidity measured as the ratio of the market capitalization of the equity traded in the company's stock exchange, deal activity measured as the number of VC deals in the country and the legality index. The table reports averages across our sample period 1990–2010 by country. Each of these variables is observed in the calendar year before the year of the exit. The legality index is constructed as in Berkowitz et al. (2003) as the weighted average of the indicators in La Porta et al. (1998): $\text{Legality index} = 0.381 * (\text{Efficiency of Judiciary}) + 0.5778 * (\text{Rule of Law}) + 0.5031 * (\text{Corruption}) + 0.3468 * (\text{Risk of Expropriation}) + 0.3842 * (\text{Risk of Contract Repudiation})$.

(*Company Size*) is measured as the logarithm of company value post-financing (i.e., immediately after the VC investment).

The geographic locations of the portfolio companies are grouped into five regions: UK & Ireland, US & Canada (North America), Continental Europe, Latin America, and Asia & Africa. The latter four cross-border regions are modelled using four regional dummy variables (one for each of the four regions treating UK & Ireland as the base).

We include four measures of domestic stock market activity and economic conditions: first, the Morgan Stanley Capital International (MSCI) country-specific market-to-book ratio for a country stock market(s), second, market liquidity measured as the total market capitalization of shares listed on a country stock market divided by gross domestic product; third, deal activity measured as the number of VC deals in the country of the portfolio company, and fourth, the (natural logarithm of the) gross domestic product (GDP) measured in £ millions at constant 2006 prices. Information on the MSCI market-to-book ratio is collected from Bloomberg; data on market liquidity and gross domestic product are collected

⁵ To clarify, we exclude VC investments that arise as a result of buyout (MBO), i.e. VC entry through buyout. However, we do include investments that are exited through buyout (MBO).

Table 3
Summary statistics.

Variables	Full sample							
	Mean	Median	STD	25th Percentile	75th Percentile	95th Percentile	Max.	Min.
<i>VC related variables</i>								
Time to exit (years)	8.35	8.00	3.24	6.00	11.01	12.00	14.00	2.00
VC Age (years)	20.17	18.00	17.25	7.00	26.00	57.00	72.00 ^a	1.00
VC Size (£m)	192.98	24.47	388.80	6.26	128.99	191.42	239.67	0.161
Fund Age (years)	4.24	3.99	2.89	1.52	6.00	9.67	11.99	0.011
Size Inv (£m)	7.38	3.07	27.100	1.18	6.37	22.79	89.13	0.155
Synd (#)	3.22	2.00	2.26	1.00	4.00	6.00	19.00	1.00
<i>Portfolio company related variables</i>								
Company Age (years)	9.22	7.00	14.95	2.00	10.00	25.00	41.00	1.00
Company Size (£m)	42.10	36.50	11.57	36.50	47.50	54.92	82.99	0.75
<i>Market related variables</i>								
Market to book	2.38	2.41	0.52	2.07	2.62	3.25	3.61	1.00
Market Liquidity	1.56	1.38	0.95	0.86	2.00	3.31	7.55	0.07
Deal Activity (#)	487.51	322.00	318.96	263.00	648.00	3961	4003	1.00
Legality Index	20.09	20.41	1.76	20.41	20.85	21.13	21.91	9.39
GDP per capita (\$)	34741	31481.7	12245.3	25121.0	39682.4	48070.4	48407.08	1327
<i>Logarithmic transformation</i>								
VC Age	2.69	2.94	0.91	2.08	3.30	4.06	4.28	0.00
VC Size	3.51	3.24	1.94	1.98	4.87	4.90	5.48	−1.83
Fund Age	1.48	1.61	0.63	0.92	1.95	2.37	2.48	−4.61
Size Inv	1.48	1.40	0.94	0.78	2.00	3.17	4.49	−3.51
Company Age	1.91	2.08	0.85	1.10	2.40	3.26	3.71	0.00
Company Size	3.73	3.62	2.39	3.62	3.88	4.02	4.42	−0.87
Deal Activity	5.91	5.78	0.88	5.58	6.48	7.96	8.29	0.00
GDP per capita	10.45	10.17	2.78	10.1	10.58	10.78	10.78	7.19

This table shows the mean, median, standard deviation, 25th, 75th and 95th percentiles, and maximum and minimum values of the continuous variables for our sample of 4502 VC investments in the period 1990–2010. Time to exit is a measure of time between investment date and date of exit in years. VC Age is measured as the difference (in years) between the founding date and the date of investment. VC Size is the size of the VC firm measured by the amount (in millions of 2006 GBP) invested by the VC firm in all its portfolio companies in the calendar year before the year of exit. Fund age is measured as the difference in years between the VC funds first investment (in any company) and the date of exit from the portfolio company for which we observe an exit. Size Inv is the amount invested in millions of 2006 GBP by the VC provider in the given portfolio company for which we observe an exit. Synd is the number of VC firms syndicated in financing the portfolio company for which we observe time to exit. Company Age is the age of the portfolio company measured as the difference (in years) between the founding date and the date of the VC investment. Company Size is the book value of assets of the portfolio company after the VC investment (in millions of 2006 GBP). Market to book is the market-to-book ratio (measured as the market value divided by the book value of equity) of the stock exchange in the country of origin of the portfolio company. Market Liquidity is market liquidity in the country of origin of the portfolio company measured as the ratio of the total market capitalization of equity traded on the company's stock exchange, divided by its gross domestic product. Deal Activity is the number of VC deals in the country of origin of the portfolio company in the calendar year before the year of exit. Legality Index is the index of the quality of the legal system in the country of origin of the portfolio company (Berkowitz et al., 2003).

^a This is Martin Currie Investment Management LTD, founded in 1881.

from the World Bank online database. We also include the legality index of the country of origin, constructed as in Cumming et al. (2006).

We also control for year effects. In the analyses reported below, we include dummy variables for the year of exit. Our alternative approach of including dummies for the deal year leaves our results qualitatively unchanged.⁶ Finally, we control for industry effects using the 12 industrial sectors of the Fama–French classification.

4. Analysis and results

4.1. Descriptive statistics

Table 2 provides information on the market and economic variables and on the legality index of the countries of origin of the portfolio companies. The table reports averages across our sample period 1990–2010 by country. The country that clearly stands out in terms of each of the four variables is the U.S. It has the highest values of market valuations (market-to-book ratio MTB), market liquidity and the number of VC deals (Deal Activity). The UK is second in terms of MTB and liquidity, but is banished to third place by Canada in terms of deal activity. In terms of the quality of their legal systems, the UK, U.S. and Canada rank only in upper 20–30 percentiles, being outranked by several European countries with Switzerland and Norway in the lead.

Table 3 reports descriptive statistics of the variables used throughout our exit analysis based on the filtered sample of 4502 VC investment observations.⁷ The average time to exit from a portfolio company is 7 years. Three out of four investments are exited within 9 years of the investment. The average age of the VC firms (VC Age) at the time of investment is around 20 years, with a median of nearly 18 years.⁸ The mean age of UK VC firms is comparable to that reported for US VC firms (Giot and Schwienbacher, 2007). The average VC Size, in terms of the mean amount invested by a VC firm, is approximately £193 million annually, with a much lower median of £24.47 million.

At the time of exit, the VC funds (in contrast to the VC firms managing the funds) have a mean age (Fund Age) of 4.24 years. As funds have a typical lifespan of 10 years, we might expect an average fund age of five years if exits were uniformly distributed over the funds lifespan. Approximately 95 percent of the funds have an age less than 9.67 years. On average, VC providers commit £7.38 million to each portfolio company (Size Inv). The distribution is highly skewed, with a minimum of just £30,000 and a maximum of £893 million. In fact, one in two portfolio companies receive £3 million (the median) or less. As in the US, syndication is common in the UK VC industry. The mean syndicate size (Synd) is three. Half of all the investments have two VC firms in the syndicate (the

⁷ For variables expressed in logarithms in our analysis, Table 2 reports statistics for both the logarithmic transformations and the non-logarithmic values.

⁸ The values used to calculate the statistics of the VC characteristics for the syndicated deals are the averages of the characteristics of the syndicate members.

⁶ These results are available on request.

Table 4
Continuous variables by region and exit route.

Variables	N	Time to exit (Years)	VC Age (Years)	VC Size (£m)	Fund Age (Years)	Size Inv (M)	Syndicate Size (#)	Company Age (years)	Company Size (£m)	Market to book	Market Liquidity	Deal Activity (#)	Legality index
<i>Breakdown of VC investments by exit route and country of portfolio company</i>													
M&A	3163	8.47	20.08	196.81	4.07	6.01	3.31	8.87	43.82	2.44	1.54	4799.56	20.16
IPO	536	6.89	21.45	209.08	4.66	13.43	3.89	11.63	47.85	2.43	1.89	2488.14	19.90
Liquidation	175	7.56	19.67	339.23	3.40	5.24	3.09	8.19	48.09	2.75	1.83	1958.21	20.57
MBO	159	7.38	30.89	254.23	3.69	18.15	3.19	11.77	44.48	2.37	1.46	1709.36	18.74
No Exit	469	–	15.86	73.42	5.47	6.92	3.17	8.35	43.34	1.79	1.41	1793.41	20.06
<i>Geographic region of portfolio company</i>													
UK & Ireland	2112	7.93	17.72	121.37	3.72	15.32	3.16	9.37	44.51	2.39	1.60	606.83	20.36
North America	1304	8.93	16.16	265.71	5.07	15.27	4.59	7.45	45.40	2.34	2.11	5830.92	20.89
Continental Europe	689	8.73	32.28	439.35	4.44	9.33	3.28	10.72	43.36	2.43	0.85	620.70	20.18
Latin America	45	7.83	18.40	174.11	4.14	5.94	2.91	14.31	37.46	2.17	0.09	455.53	15.76
Asia & Africa	352	7.31	26.27	159.81	3.93	6.29	2.79	11.33	43.45	2.37	0.86	668.29	15.81
<i>Financing stages</i>													
Early stage	1756	8.84	22.30	162.77	3.11	7.96	5.21	5.41	10.41	2.41	1.49	2038.12	20.85
Expansion	2272	7.91	19.24	236.41	4.22	10.73	3.41	9.61	47.34	2.38	1.91	2505.31	20.74
Later stage	474	6.21	18.18	273.45	4.88	9.29	3.02	12.9	63.99	2.21	2.38	3614.12	20.14

This table shows the mean values of the continuous variables by exit route and geographic region of the portfolio company. *VC Age* is measured as the difference (in years) between the founding date and the date of investment. *VC Size* is the size of the VC firm measured by the amount (in millions of 2006 GBP) invested by the VC firm in all its portfolio companies in the calendar year before the year of exit. *Fund age* is measured as the difference in years between the VC funds first investment (in any company) and the date of exit from the portfolio company for which we observe an exit. *Size Inv* is the amount invested in millions of 2006 GBP by the VC provider in the given portfolio company for which we observe an exit. *Synd* is the number of VC firms syndicated in financing the portfolio company for which we observe time to exit. *Company Age* is the age of the portfolio company measured as the difference (in years) between the founding date and the date of the VC investment. *Company Size* is the book value of assets of the portfolio company after the VC investment (in millions of 2006 GBP). *Market to book* is the market-to-book ratio (measured as the market value divided by the book value of equity) of the stock exchange in the country of origin of the portfolio company. *Market Liquidity* is the market liquidity in the country of origin of the portfolio company, measured as the ratio of the total market capitalization of equity traded on the company's stock exchange divided by its gross domestic product. *Deal Activity* is the number of VC deals in the country of origin of the portfolio company in the calendar year before the year of exit. *Legality Index* is the (logarithm of the) index of the quality of the legal system in the country of origin of the portfolio company (Berkowitz et al., 2003).

Table 5

Types of exit routes.

Variables	N	Number of observations by exit route					Number of observations by financing stage		
		M&A Exit	IPO Exit	Liquidation	MBO	No exit	Early stage	Expansion stage	Later stage
Breakdown of VC investments by geographic region of portfolio company									
UK & Ireland	2112	1602 (76%)	164 (8%)	43 (2%)	61 (3%)	242 (11%)	863 (41%)	1123 (53%)	126 (6%)
North America	1304	822 (63%)	203 (16%)	108 (8%)	33 (3%)	138 (11%)	494 (38%)	552 (42%)	258 (20%)
Continental Europe	689	507 (74%)	96 (14%)	21 (3%)	19 (3%)	46 (7%)	237 (34%)	407 (59%)	45 (7%)
Latin America	45	31 (69%)	7 (16%)	0	2 (4%)	5 (11%)	16 (36%)	23 (51%)	6 (13%)
Asia & Africa	352	201 (57%)	66 (19%)	3 (1%)	44 (13%)	38 (11%)	146 (41%)	167 (47%)	39 (11%)
Total	4502	3163	536	175	159	469	1756	2272	474

This table shows the proportion of investments that are exited via M&As, IPOs, liquidations and MBOs between 1990 and 2010 by origin (region) of portfolio company and exit route. Figures in brackets are the different types of exits as proportions of the total number of investment by region. The table also shows the number of observations by financing stage.

Table 6

Nonparametric estimation of exit rates.

Exit method by geographical region	No of obs.	Years after investment											
		1	2	3	4	5	6	7	8	9	10	11	12
<i>UK & Ireland</i>	2112												
M&A		0.019	0.067	0.127	0.217	0.320	0.433	0.561	0.717	0.858	0.946	0.974	0.988
IPO		0.033	0.098	0.197	0.230	0.459	0.541	0.574	0.754	0.853	0.984	1.000	1.000
Liquidation		0.023	0.093	0.163	0.233	0.302	0.372	0.558	0.837	0.884	0.930	0.954	1.000
MBO		0.018	0.067	0.152	0.226	0.323	0.470	0.573	0.707	0.842	0.902	0.939	0.976
<i>North America</i>	1304												
M&A		0.016	0.039	0.074	0.141	0.226	0.298	0.429	0.577	0.724	0.849	0.919	0.972
IPO		0.061	0.182	0.212	0.394	0.515	0.606	0.636	0.727	0.970	1.000	0.000	0.000
Liquidation		0.056	0.093	0.157	0.222	0.259	0.380	0.509	0.630	0.778	0.944	0.963	0.000
MBO		0.005	0.049	0.084	0.182	0.276	0.414	0.503	0.591	0.714	0.783	0.877	0.936
<i>Continental Europe</i>	689												
M&A		0.010	0.022	0.055	0.122	0.201	0.335	0.468	0.588	0.791	0.919	0.974	0.988
IPO		0.021	0.083	0.135	0.208	0.344	0.458	0.604	0.719	0.854	0.927	0.979	1.000
Liquidation		0.000	0.048	0.095	0.191	0.238	0.286	0.381	0.762	0.905	0.952	0.952	1.000
MBO		0.000	0.000	0.158	0.211	0.316	0.474	0.632	0.842	0.947	1.000	0.000	0.000
<i>Latin America</i>	45												
M&A		0.032	0.032	0.097	0.129	0.258	0.452	0.613	0.774	0.871	0.936	0.968	1.000
IPO		0.143	0.143	0.143	0.286	0.429	0.429	0.429	0.571	0.714	1.000	0.000	0.000
Liquidation		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
MBO		0.000	0.000	0.000	0.000	0.000	0.500	0.000	0.000	0.000	1.000	0.000	0.000
<i>Asia & Africa</i>	352												
M&A		0.010	0.105	0.199	0.269	0.398	0.493	0.607	0.786	0.881	0.955	0.975	0.985
IPO		0.091	0.152	0.258	0.333	0.432	0.546	0.682	0.788	0.864	0.909	0.955	0.970
Liquidation		0.000	0.000	0.000	0.000	0.667	0.000	0.000	1.000	0.000	0.000	0.000	0.000
MBO		0.000	0.068	0.182	0.364	0.409	0.591	0.750	0.864	0.955	1.000	0.000	0.000

The cumulative exit rates are estimated using the Kaplan–Meier nonparametric method. The table shows the breakdown of exit rates by country of portfolio company and exit method. The exit times are measured between the investment date and date of exit, in years. 0 indicates that no exit has taken place by the end of a given year post-investment. 1 indicates that 100% of the investments have been exited.

median of *Synd*) and one in twenty investments involve more than five syndicated investors (based on the 95th percentile of *Synd*).

The average age of a portfolio company at the time of investment is over 9 years, but company age is highly variable with a standard deviation of 15 years. The mean size (book value of assets) of a portfolio company is close to £24 million post-investment. The smallest company size is £42,000 while, for the largest company in our sample, the book value of assets is around £850 million. On average, the country market-to-book ratio is 2.38 and the stock market liquidity is 1.56 across all regions. On average, there are 492 VC deals per calendar year in the country of origin of the portfolio company. The mean value of the legality index is 20.09, with a minimum of 9.39 and a maximum of 21.91 (see also Table 2 for legality by country).

Table 4 gives a breakdown of time to exit and the other variables by exit route, portfolio company region and financing stage of the VC investment.

As shown in the table, the most widely used exit route for VC investments in our sample is the M&A (including trade sales),

followed by the IPO, liquidation, and finally the MBO. We find that 3163 investments in our sample are exited through M&A, 536 through IPO, 175 through liquidation, and 159 through MBO. We also observe 469 investments reported by VC firms as not (yet) exited as of the end of 2010.

In terms of the average time to exit by exit route, IPO is the quickest exit route (6.07 years) followed by MBO (6.98 years), M&A (7.12 years) and liquidation (7.55 years). Examining the average time to exit by region, we find slight variations with average time to exit ranging from seven years in Africa & Asia to eight years in the UK & Ireland and in Latin America, to just under nine years in Continental Europe and North America. None of the regional differences in average exit time are statistically significant.

Examining the time to exit by financing stage, we find that early-stage VC investments on average take longest to exit (nearly 9 years) while later-stage investments are exited most quickly (after just over 6 years). The average time to exit from expansion-stage investments lies between these two figures at just under 8 years. The average times to exit we report across all the

Table 7a

Estimation of time to exit (cross-border effect).

Variables	Model I				Model II			
	M&A exits		IPO exits		M&A exits		IPO exits	
	Coef.	p-Value	Coef.	p-Value	Coef.	p-Value	Coef.	p-Value
Cross-border	−0.697***	(0.000)	−0.294***	(0.004)	–	–	–	–
North America	–	–	–	–	−1.286***	(0.000)	−0.369***	(0.012)
Cont. Europe	–	–	–	–	−0.063	(0.292)	0.137	(0.278)
Latin America	–	–	–	–	−0.045	(0.823)	−0.189	(0.468)
Asia & Africa	–	–	–	–	−0.002	(0.979)	0.282*	(0.092)
<i>VC related variables</i>								
VC age	0.028	(0.439)	0.128*	(0.094)	0.031	(0.397)	0.138*	(0.072)
Old VC	−0.047**	(0.037)	−0.104***	(0.009)	−0.049**	(0.032)	−0.121***	(0.001)
VC size	−0.001	(0.571)	0.001	(0.588)	−0.001	(0.671)	−0.001	(0.814)
Fund age	0.022	(0.656)	−0.04	(0.706)	0.025	(0.602)	−0.010	(0.922)
Old fund	−0.016	(0.639)	−0.049	(0.370)	−0.017	(0.602)	−0.051	(0.316)
Size inv	−0.043**	(0.033)	−0.148***	(0.007)	−0.048**	(0.017)	−0.161***	(0.003)
TOP25% size inv	−0.005	(0.403)	0.040**	(0.022)	−0.005	(0.425)	0.044***	(0.004)
Synd	−0.022***	(0.002)	−0.019**	(0.050)	−0.016**	(0.024)	−0.014*	(0.085)
Old synd	−0.013*	(0.100)	0.015	(0.175)	−0.013*	(0.095)	0.015	(0.179)
<i>Portfolio company related variables</i>								
Company Age	−0.022	(0.326)	0.045	(0.351)	−0.023	(0.317)	0.046	(0.364)
Company size	0.052	(0.218)	−0.054	(0.478)	0.055	(0.194)	−0.055	(0.498)
Constant	−6.532***	(0.000)	−4.023***	(0.000)	−6.573***	(0.000)	−3.926***	(0.000)
Pseudo R square	0.155		0.111		0.168		0.131	
No. of exit events	3163		536		3163		536	
No. of censored/unexcited	469		469		469		469	

The coefficients of the models are estimated through maximum likelihood estimation. The values in brackets are *p*-values based on clustered standard errors to correct for within-group correlations. *VC Age* is the logarithm of VC age, *Old VC* is the logarithm of the VC age of the oldest 25% of VC firms. *VC Size* is the logarithm of the total amount committed by the VC firm in the year prior to the year of exit. *Fund age* is the logarithm of the age of the VC fund in the year of exit. *Old Fund* is the logarithm of the fund age in the case of the oldest 25% of funds. *Size Inv* is the logarithm of the amount invested by VC provider(s) in the portfolio company. *TOP25% Size Inv* is the highest 25% of *Size Inv*. *Synd* is the number of VC firms syndicated in financing the portfolio company. *Old Synd* captures the presence of an old fund in the syndicate. *Company Age* is the logarithm of portfolio company age, measured in years from the founding date to the date of VC investment. *Company Size* is the logarithm of the book value of assets of the portfolio company after the VC investment. *MTB* is the market-to-book ratio for the stock exchange in the country of origin of the portfolio company. *Market Liquidity* in the country of origin of the portfolio company is measured as the ratio of the total market capitalization of equity traded on the company's stock exchange divided by its gross domestic product. *Deal Activity* is the (logarithm of) the number of deals in a deal year for a given country. *Legality* is the weighted average (based on Berkowitz et al., 2003) of the indicators for the efficiency of the judicial system, the rule of law, corruption, the risk of expropriation and the risk of contract repudiation (La Porta et al., 1998). *North America* is a dummy variable for that region of origin, and similarly for the other regions. We include dummies for stage of financing, industry and year of exit. ***, ** and * indicate significance levels of 1%, 5% and 10% respectively.

regions are higher than those reported in some previous studies for US investments alone. For instance, Giot and Schwienbacher (2007) report a time to exit of 4.6 years for early-stage, 3.4 years for expansion-stage and 3 years for later-stage investments. Part of this difference is due to the comparatively longer times to exit during the period 2007–2010. For the rest of our sample period 1990–2006, we find the average time to exit to be only slightly higher than that reported by Giot and Schwienbacher. Specifically, it is 5.85 years for early-stage, 4.90 years for expansion-stage and 3.92 years for later-stage investments. The longer times to exit between 2007 and 2010 are likely due to adverse market conditions. Consequently, our multivariate analysis controls for market conditions and year effects.

Table 4 also provides a univariate analysis of exit routes in terms of the associated characteristics of VC backers, deals, portfolio companies and macro factors. The subsample of IPO exits has, on average, the largest syndicates and the oldest portfolio companies. The liquidation subsample, on the other hand, has the smallest average size of investment and the youngest portfolio companies. Markets with higher liquidity have a higher number of IPO exits and liquidations. The non-exited investments are in portfolio companies that are smaller on average at the time of investment than the exited ones.

In terms of the impact of geographical region, UK & Ireland and North America have the highest average amounts invested (*Size Inv*), followed by Continental Europe. At the time of exit, stock market liquidity is highest in North America followed by the UK & Ireland. The oldest portfolio companies are based in Latin America

and Asia & Africa. These findings suggest that VC firms invest more in countries with well-developed stock markets and sound legal systems. When investing in countries where the stock markets are less developed and the legal regime provides investors with less protection, they target mature companies.

Table 5 shows the numbers (and proportions) of investments by exit route and financing stage, by region. In terms of the relative numbers of investments by region, the number of investments by UK VC firms in their domestic region, the UK & Ireland, is just under half (47 percent) of the total number of investments. This is followed by North America with 29 percent of investments, and Continental Europe with 15 percent. The largest number and proportion of investments occurs in the expansion stage (2272 investments accounting for just over 50 percent of the total) followed by early-stage investments (39 percent), while later stages investments are less common (11 percent). For all regions, exits through M&A are more frequent than through other routes. Nearly 75 percent of the VC firms in the UK & Ireland and Continental Europe exit their portfolio companies through M&A, as compared to 63 percent in North America and 57 percent in Asia & Africa. Nearly 19 percent of exits in Asia & Africa are through IPOs, followed by North America (16 percent), Continental Europe (14 percent) and the UK & Ireland (8 percent). VC firms investing in North America liquidate more (and a larger proportion of) investments than those investing elsewhere.

Table 6 reports the cumulative exit rates by post-investment year. The exit rates are estimated as 'failure rates' using the non-parametric Kaplan–Meier method shown in Eq. (1) above.

Table 7b

Estimation of time to exit (North American effect).

Variables	Model III				Model IV				Model V			
	M&A exits Coef.	p-Val	IPO exits Coef.	p-Val	M&A exits Coef.	p-Val	IPO exits Coef.	p-Val	M&A exits Coef.	p-Val	IPO exits Coef.	p-Val
North America	−1.333***	(0.000)	−0.449***	(0.001)	−0.427	(0.227)	−0.371	(0.574)	−0.403	(0.477)	−0.317	(0.546)
Legality	0.0071	(0.2150)	−0.054**	(0.015)	−0.093	(0.145)	−0.062**	(0.015)	−0.071	(0.016)	−0.051***	(0.004)
Market liquidity	0.388***	(0.000)	−0.092**	(0.034)	0.664***	(0.000)	−0.169**	(0.041)	0.524***	(0.000)	−0.152**	(0.010)
MTB	−0.099**	(0.024)	0.148	(0.150)	−0.045*	(0.074)	0.212*	(0.095)	−0.096*	(0.070)	0.184	(0.178)
Deal activity	0.135***	(0.000)	−0.011	(0.821)	0.150***	(0.000)	−0.059	(0.379)	0.181***	(0.000)	−0.045	(0.252)
GDP per capita	0.249***	(0.000)	0.023*	(0.072)	0.280***	(0.000)	0.030*	(0.069)	0.293***	(0.000)	0.048*	(0.054)
Market liquidity × North America	–	–	–	–	−0.0703	(0.250)	0.047	(0.757)	–	–	0.078*	(0.073)
MTB × North America	–	–	–	–	−0.082*	(0.095)	−0.106	(0.309)	−0.096*	(0.051)	–	–
Deal Activity × North America	–	–	–	–	−0.041	(0.637)	0.081	(0.386)	–	–	–	–
GDP per capita × North America	–	–	–	–	−0.471**	(0.011)	0.041	(0.161)	−0.523***	(0.000)	0.053*	(0.092)
<i>VC related variables</i>												
VC Age	0.038	(0.294)	0.105	(0.188)	0.044	(0.227)	0.131*	(0.082)	0.055	(0.136)	0.109*	(0.097)
Old VC	−0.059**	(0.011)	−0.116***	(0.002)	−0.055**	(0.016)	−0.137***	(0.000)	−0.060***	(0.010)	−0.122***	(0.002)
VC Size	−0.004**	(0.015)	0.001	(0.673)	−0.005***	(0.002)	0.001	(0.542)	−0.005***	(0.002)	0.001	(0.622)
Fund Age	0.007	(0.899)	−0.006	(0.960)	0.001	(0.979)	−0.022	(0.868)	0.015	(0.210)	−0.017	(0.881)
Old Fund	−0.019	(0.566)	−0.059	(0.354)	−0.022	(0.520)	−0.038	(0.557)	−0.047	(0.166)	−0.055	(0.401)
Size Inv	−0.033	(0.104)	−0.179***	(0.000)	−0.021	(0.301)	−0.187***	(0.000)	−0.018	(0.378)	−0.184***	(0.000)
TOP25% Size Inv	−0.005	(0.483)	0.045***	(0.003)	−0.005	(0.459)	0.047***	(0.001)	−0.007	(0.297)	0.046***	(0.002)
Synd	−0.028***	(0.000)	−0.013*	(0.078)	−0.033***	(0.000)	−0.014*	(0.087)	−0.027***	(0.000)	−0.011*	(0.099)
Old Synd	−0.004	(0.643)	0.019*	(0.098)	0.001	(0.918)	0.017	(0.132)	0.001	(0.926)	0.018	(0.109)
<i>Portfolio company related Variables</i>												
Company Age	0.006	(0.811)	0.06	(0.215)	0.008	(0.737)	0.057	(0.264)	0.006	(0.797)	0.065	(0.174)
Company Size	0.099**	(0.032)	−0.067	(0.412)	0.073*	(0.094)	−0.067	(0.444)	0.091*	(0.059)	−0.067	(0.409)
Constant	−8.805***	(0.000)	−3.206**	(0.020)	−8.689***	(0.000)	−3.150**	(0.030)	−7.849***	(0.000)	−3.203**	(0.015)
Pseudo R square	0.167		0.133		0.182		0.141		0.174		0.138	
No. of exit events	3163		536		3163		536		3163		536	
No. of censored/unexited	469		469		469		469		469		469	

The coefficients of the models are estimated through maximum likelihood estimation. The values in brackets are p-values based on clustered standard errors to correct for within-group correlations. VC Age is the logarithm of VC age, Old VC is the logarithm of the VC age in the case of the oldest 25% of VC firms. VC Size is the logarithm of the total amount committed by the VC firm in the year prior to the year of exit. Fund age is the logarithm of the age of the VC fund in the year of exit. Old Fund is the logarithm of the fund age in the case of the oldest 25% of funds. Size Inv is the logarithm of the amount invested by VC provider(s) in the portfolio company. TOP25% Size Inv is the highest 25% of Size Inv. Synd is the number of VC firms syndicated in financing the portfolio company. Old Synd captures the presence of an old fund in the syndicate. Company Age is the logarithm of the portfolio company's age, measured in years from the founding date to the date of VC investment. Company Size is the logarithm of the book value of assets of the portfolio company after the VC investment. MTB is the market-to-book ratio for the stock exchange in the country of origin of the portfolio company. Market Liquidity in the country of origin of the portfolio company is measured as the ratio of the total market capitalization of equity traded on the company's stock exchange divided by its gross domestic product. Deal Activity is the (logarithm of) the number of deals in the deal year for a given country. Legality is the weighted average (based on Berkowitz et al., 2003) of the indicators of the efficiency of the judicial system, the rule of law, corruption, the risk of expropriation and the risk of contract repudiation (La Porta et al., 1998). North America is a dummy variable for that region of origin. We include dummies for stage of financing, industry and year of exit. We exclude other region dummies from the regression and the chi-square test is 3.68 (0.301) for M&As and 3.07 (0.250) for IPOs, which are statistically insignificant. ***, ** and * indicate significance levels of 1%, 5% and 10% respectively. The joint test of the interaction terms for the IPO exits is significant at 5% with Chi-square value of 8.14 and p-value (0.024).

Table A1
Correlation matrix.

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1 LnVC age	1																			
2 Old VC	0.489**	1																		
3 VC Size	0.425**	0.412**	1																	
4 Fund Age	0.207*	0.123	0.211*	1																
5 Old Fund	0.088	0.024	0.111	0.499**	1															
6 Size Inv	0.083	0.119	0.384**	0.133	0.054	1														
7 TOP25%Size Inv	0.077	0.103	0.308**	0.029	0.001	0.482**	1													
8 Synd	0.023	0.004	0.012	0.281*	0.223*	0.159	0.018	1												
9 Old Synd	0.055	0.059	0.051	0.192	0.469**	0.076	0.025	0.446**	1											
10 Company Age	0.048	0.025	0.034	0.027	0.019	0.047	0.113	0.13	0.081	1										
11 Company Size	0.012	0.020	0.025	0.019	0.009	0.056	0.022	0.025	0.035	0.058	1									
12 MTB	0.003	0.032	0.029	0.146	0.149	0.160	0.052	0.407**	0.334**	0.081	0.017	1								
13 Market Liquid	0.045	0.067	0.185	0.242*	0.059	0.006	0.039	0.090	0.040	0.014	0.005	0.064	1							
14 Deal Activity	0.005	0.066	0.231*	0.137	0.102	0.046	0.028	0.147	0.178	0.067	0.001	0.190	0.427**	1						
15 Legality	0.049	0.077	0.077	0.040	0.044	0.161	0.166	0.164	0.091	0.098	0.018	0.264*	0.025	0.003	1					
16 GDP per capita	−0.081	−0.118	−0.148	0.141	0.065	−0.112	−0.141	0.195	0.095	−0.081	0.004	0.190	−0.115	−0.061	0.313**	1				
17 North America	0.072	0.090	0.003	0.183	0.177	0.093	0.004	0.419**	0.330**	0.078	0.001	0.429**	0.056	0.201*	0.292*	0.482**	1			
18 Cont. Europe	0.215*	0.228*	0.287*	0.038	0.008	0.044	0.044	0.041	0.044	0.032	0.022	0.197	0.047	0.101	0.021	−0.258	0.272*	1		
19 Latin America	0.021	0.030	0.016	0.013	0.008	0.038	0.047	0.029	0.005	0.037	0.055	0.079	0.044	0.029	0.243*	−0.132	0.059	0.039	1	
20 Asia & Africa	0.056	0.083	0.104	0.035	0.027	0.163	0.137	0.083	0.042	0.065	0.007	0.140	0.004	0.017	0.480**	−0.279	0.185	0.123	0.027	1

This table shows the correlations between the variables. ***, ** and * indicate the 1%, 5% and 10% significance levels.

The figures are broken down by both region and exit route. Focusing on IPO exits, we find that more than half the IPO exits (51.5 percent) in North America have taken place by the end of the fifth post-investment year. By the end of year five, the rate of exit through IPO is highest in North America, followed by the UK & Ireland, and lowest in Continental Europe. The cumulative exit rate is lower for M&A exits than for IPOs in all regions over the first five years post-investment. It takes almost seven years for half of the M&A exits to occur in the UK & Ireland, Latin America and in Asia & Africa, and up to eight years in North America and Continental Europe. We observe comparatively high cumulative exit rates in Asia & Africa. In particular, M&A exit rates are highest in Asia & Africa until the end of year six followed by the UK & Ireland and North America. By contrast, Continental Europe is characterised by comparatively low exit rates for both M&A and IPO.

Overall, the results show that, in most regions, UK VC firms first try to exit their portfolio companies through IPO, then by M&A or MBO. Only if all else fails, VC firms resort to exit through liquidation. The exception is in Asia & Africa, where exits through liquidation occur comparatively quickly during the first five years following investment. These initial, univariate results suggest several region-specific exit effects that we explore next in our multivariate analysis.

4.2. Multivariate analysis

We conduct a multivariate analysis to explore whether there is a systematic difference between domestic and cross-border investments controlling for the characteristics of VC backers, deals and portfolio companies. Table 7 reports the results of a parametric Accelerated Failure Time (AFT) frailty model of M&A and IPO exits. The model corrects for frailty in the sense of heterogeneity due to unobserved individual VC firm effects. Since the data span many VC firms, we control for cross-correlation among the investments of a given VC firm using clustered standard errors. Failure to control for such effects might lead to biased estimates.

The results of Model I in Table 7 suggest a significant and pronounced difference between cross-border and domestic investments in the time to exit for both M&A and IPO. The negative coefficients of the cross-border indicator in both the M&A and the IPO equation indicate that cross-border investments are exited more quickly than domestic investments through either route. To quantify the impact of cross-border investments on the time to exit, we derive the time ratio as the exponential of the coefficients of the cross-border coefficients (Espenlaub et al., 2012). The time ratios are $\exp(-0.697) = 0.50$ for M&A exits and $\exp(-0.294) = 0.75$ for IPO exits. The M&A time ratio of 0.50 indicates that, all else equal, the time to M&A exits from cross-border investments is half that from domestic investments. The IPO time ratio shows that IPO exits from cross-border investments happen within three quarters of the time taken to exit domestic investments, all else being equal.

The strength of this apparent 'cross-border effect' seems striking, and the next step in our analysis is to explore the precise origin of this effect. First, we seek to understand whether there are variations across the regions that make up the cross-border subsample. Model II re-estimates Model I using dummy variables for each of the four regions (outside the home region of the UK & Ireland). None of the coefficients of regional dummies are significant with the exception of the North American dummy. This means there are no significant differences in exit times between the UK & Ireland and any other regions except for North America. All else being equal, average times to exit in all other cross-border regions (besides North America) are the same as those in the UK & Ireland. What seems to be a 'cross-border effect' in exit times in Model I turns out to be driven by the distinctive characteristics and the predominance of the North American investments within our

Table A2

This table shows the results of the cross-border and regional effects controlling for market and legal system variables. The variables are as defined in Table 1.

Variables	Model A I	Model A II	Model A III		
	M&A Exits	IPO Exits	M&A Exits	IPO Exits	IPO vs. M&A
Cross-border	−0.885*** (0.000)	−0.354*** (0.002)	–	–	–
North America	–	–	−1.032*** (0.000)	−0.408*** (0.003)	0.470*** (0.000)
Cont. Europe	–	–	−0.027 (0.694)	0.249 (0.140)	0.177 (0.272)
Latin America	–	–	−0.057 (0.156)	−0.047 (0.924)	0.0247 (0.665)
Asia & Africa	–	–	−0.1649 (0.124)	0.085 (0.089)	0.118 (0.333)
Legality	−0.034* (0.065)	−0.064*** (0.004)	0.005 (0.789)	−0.043* (0.071)	0.047* (0.086)
Market Liquidity	0.313*** (0.000)	−0.074* (0.080)	0.384*** (0.000)	−0.096** (0.041)	0.224*** (0.000)
MTB	−0.162*** (0.000)	0.139 (0.182)	−0.104** (0.021)	0.159 (0.111)	−0.0519*** (0.000)
Deal Activity	0.163*** (0.000)	−0.004 (0.998)	0.135*** (0.000)	−0.019 (0.676)	−0.144** (0.017)
GDP per capita	0.256*** (0.000)	0.091 (0.228)	0.236** (0.015)	0.277* (0.098)	−0.131** (0.078)
VC Age	0.046 (0.207)	0.112 (0.166)	0.042 (0.256)	0.097 (0.227)	0.011 (0.879)
Old VC	−0.062*** (0.007)	−0.115*** (0.002)	−0.060*** (0.010)	−0.117*** (0.002)	0.001 (0.985)
VC Size	−0.005*** (0.002)	0.001 (0.861)	−0.004** (0.024)	−0.001 (0.934)	0.002 (0.116)
Fund Age	0.100* (0.057)	0.015 (0.897)	0.008 (0.887)	−0.059 (0.631)	0.0269 (0.132)
Old Fund	−0.053 (0.120)	−0.068 (0.282)	−0.023 (0.499)	−0.038 (0.558)	−0.018** (0.077)
Size Inv	−0.025 (0.231)	−0.177*** (0.001)	−0.038* (0.065)	−0.188*** (0.000)	0.032 (0.621)
TOP25% Size Inv	−0.006 (0.380)	0.045*** (0.003)	−0.005 (0.480)	0.049*** (0.001)	0.027 (0.233)
Synd	−0.030*** (0.000)	−0.013* (0.056)	−0.028*** (0.000)	−0.015* (0.090)	−0.122*** (0.000)
Old Synd	0.001 (0.868)	0.019* (0.091)	−0.004 (0.593)	0.018 (0.114)	0.079*** (0.000)
Company Age	0.005 (0.831)	0.061 (0.199)	0.005 (0.818)	0.062 (0.202)	0.098** (0.020)
Company Size	0.096** (0.038)	−0.065 (0.440)	0.101** (0.030)	−0.075 (0.383)	−0.254*** (0.003)
Constant	−4.356*** (0.000)	−1.89 (0.169)	−10.224*** (0.000)	−5.785** (0.024)	–
Obs	3163	536	3163	536	3699

sample. Our multivariate AFT results in Model II suggest that average times to exit are significantly *shorter* in North America than anywhere else, including the domestic region (UK & Ireland). The time ratio of the North American indicator in Model II is 0.29 ($\exp(-1.286)$). This shows that investments in North America are exited within less than one third of the time it takes to exit investments elsewhere in the world.

Next, we explore the impact of variations in the nature and development of the regional economies, capital markets and legal systems. In Model III of Table 7b, we include continuous variables describing the regional economies, capital markets and legal systems. Given the significance of the North American indicator in Model II, Model III includes the North American indicator to control for the impact of North American investments. North American observations both dominate our sample and are likely to be distinctive (see Table 4 for descriptive statistics by region). To assess the impact of market and macro variables on exit times worldwide (including outside North America), it is important to control for the influence of North American observations. Previous cross-country venture-capital studies note the predominance of North American observations and the need to control for this (Cumming et al., 2010; Wang and Wang, 2012). While we include the North

American indicator, we exclude the other regional dummies as these are found to insignificant in Model II.⁹ We further confirm that the regional dummies (except that for North America) are also insignificant in models that include the macro factors similar to Model III in Table 7b, by estimating the equations of Model A II in Table A2 (in the Appendix).

The results in Model III, Table 7b, show that IPO exit is significantly accelerated by greater stock market liquidity and greater effectiveness of the legal system (as indicated by a higher regional legality index). Our findings that market liquidity and legality speed up the time to IPO exit are in line with the predictions and findings of Black and Gilson (1998), Giot and Schwienbacher (2007) and Schwienbacher (2008) for liquidity and Cumming et al. (2006) for legality.

M&A exits are speeded up by higher stock market valuations as measured by the average market-to-book ratios *MTB* of the stock exchanges in the region of the portfolio company. This is in line with our expectations and consistent some of the findings of

⁹ A likelihood ratio test of the regional dummies (other than the North American dummy) shows that they are jointly insignificant (chi-square 3.68, *p*-value 0.31).

Table A3

The table shows the results of MBO and liquidation exits. The variables are as defined in Table 1.

Variables	Model A IV		Model A V	
	MBO Exits	Liquidation Exits	MBO Exits	Liquidation Exits
Cross-border	−0.1443*** (0.000)	−0.3447*** (0.000)	–	–
North America	–	–	−0.570* (0.054)	−0.632** (0.023)
Cont. Europe	–	–	0.108 (0.711)	0.211 (0.471)
Latin America	–	–	0.171 (0.847)	−0.242 (0.598)
Asia & Africa	–	–	0.301 (0.236)	−0.701 (0.262)
VC Age	0.361 (0.145)	−0.085 (0.571)	0.693*** (0.009)	0.018 (0.898)
Old VC	−0.215 (0.151)	−0.087 (0.390)	−0.312** (0.037)	−0.210** (0.026)
VC Size	0.020*** (0.009)	0.002 (0.806)	0.019** (0.024)	0.014** (0.029)
Fund Age	0.126 (0.634)	−0.920*** (0.000)	0.122 (0.650)	−1.400*** (0.000)
Old Fund	−0.117 (0.566)	0.400** (0.018)	−0.169 (0.377)	0.498*** (0.002)
Size Inv	0.043 (0.690)	0.112 (0.278)	0.047 (0.661)	−0.008 (0.934)
TOP25% Size Inv	0.030 (0.366)	−0.048* (0.095)	0.046 (0.152)	−0.016 (0.530)
Synd	−0.174** (0.021)	−0.005 (0.809)	−0.205*** (0.003)	0.011 (0.650)
Old Synd	0.145 (0.170)	−0.026 (0.281)	0.174** (0.014)	−0.029 (0.221)
Company Age	0.101 (0.379)	0.059 (0.560)	0.190* (0.084)	0.034 (0.734)
Company Size	0.365 (0.266)	0.312 (0.103)	0.231 (0.431)	−0.041 (0.796)
Constant	−1.332*** (0.000)	−1.850*** (0.000)	−1.456*** (0.000)	−1.937*** (0.000)
Obs	159	175	159	175

Cumming and Johan (2010). The impact on M&A exits of liquidity and deal activity in Model III is likely to be driven by variations across regions. North America is characterized by highly liquid stock markets and active VC markets (high deal activity) but comparatively slow M&A exits (see Tables 4 and 6). By contrast, other regions show quicker M&A exits despite relatively low market liquidity and deal activity. Thus, higher market liquidity and greater deal activity are associated with slower M&A exits across regions.

Gross domestic product (*GDP per capita*) appears to slow down the time to exit for both M&A and IPO. While the coefficients are relatively small, particularly for IPO exits, their positive sign suggests that more affluent regions are associated with slower exits than relatively poorer regions. This result is consistent with the findings of Wang and Wang (2012), who attribute the outperformance by poorer regions to the catch-up (convergence) effect discussed in the economic growth literature.

The coefficients of the North American indicator in Models II and III are almost the same, indicating that controlling for macro factors does not change this coefficient. This suggests that the distinctive characteristics of the North American economies, their legal systems and capital markets, do not explain the distinctive exit behaviour of the North American subsample that is captured by the North American indicator in Models II and III.

The continued significance of the North American indicator also suggests that in order to get reliable results on the impact of the macro variables across all the regions we may need to further

understand and control for the distinctive impact of the North American subsample. Specifically, we need to explore whether the impact of the macro factors differ between North America and the other regions (including the UK & Ireland). To this end, we include four interaction terms in Model IV: each of the four macro variables (market liquidity, market-to-book ratio, deal activity and per capital GDP) is interacted with the North America indicator. As some of the interaction terms are statistically insignificant, we estimate a more parsimonious specification (excluding insignificant interaction terms) in Model V.

In Model III–V, we find that IPO exits are speeded up by more liquid stock markets across all regions (as shown by the statistically significant negative coefficient of *Market Liquidity*). Comparing the coefficients of the three models, we find that the coefficients of the un-interacted *Market Liquidity* variable in the IPO equation becomes more negative after including the interaction terms in Models IV–V. Relative to Model III, the coefficient of *Market Liquidity* almost doubles in absolute terms (from −0.092 to −0.169) in the IPO equation of Model IV. The corresponding time ratios are 0.912 and 0.844. The ratio of 0.912 suggests that without controlling for the interaction terms, a one-unit increase in liquidity causes the time to IPO exit to be multiplied by 0.912, i.e. a one-unit increase in liquidity reduces average time to exit by 9 percent (1 − 0.912). By contrast, when we control for the interaction terms, a one-unit increase in liquidity reduces average time to IPO exit by almost 16 percent.

These results show that by failing to control for the fact that liquidity and the other macro factors affect the time to exit differently in North America, we underestimate the true accelerating effect of *Market Liquidity* on IPO exit across all the other regions outside North America. Controlling for the differential impact of the macro factors in North America in Models IV, we find that the effect of liquidity on IPO exit is in fact almost twice as strong as suggested by Model III. While the effect of the un-interacted market-to-book ratio *MTB* changes from Model III to Model IV, the results of Model III and Model V are almost the same. The other two un-interacted macro factors, *Deal Activity* and *GDP per capita* are broadly the same across the three models.

The coefficient of the un-interacted North American indicator is no longer statistically significant in Models IV and V for both M&A and IPO exits suggesting that the distinctiveness of North America operates (and is explained) largely through the distinctive impact of the macro variables in that region. To capture the full effects of a macro variable in North America, we need to add the coefficient of the interaction term of the variable with the North American indicator to the coefficient of the un-interacted macro variable.

For IPO exits, the total impact of *Market Liquidity* on time to exit in North America is given by the sum of the un-interacted *Market Liquidity* coefficient (−0.152 in Model V) and the coefficient of the interaction term of *Market Liquidity* and *North America* (0.078). The interaction term offsets nearly half the direct effect of un-interacted *Market Liquidity*. This indicates that the time taken by VC backers to exit through an IPO is substantially and significantly less sensitive to variations in liquidity in North America than elsewhere.

In Model III, we find that M&A exits are quicker in markets characterized by higher valuations in terms of market-to-book ratio (*MTB*). Models IV and V show that the sensitivity of the time to exit to changes in valuations (*MTB*) is more pronounced in North America than elsewhere. This is indicated by the significant, negative coefficients of the interaction terms of *North America* × *MTB* in the M&A equations of Models IV and V.

The results for the cross-border and regional effects are broadly similar when we examine the remaining two exit routes, namely MBOs and liquidations (see Models A IV to A VIII in Tables A3

Table A4

This table shows the results of the North American effect interacted with the market variables for MBO and liquidation exits.

Variables	Model A VI		Model A VII		Model A VIII
	MBO	Liquidation	MBO	Liquidation	MBO vs. Liquidation
	Exits	Exits	Exits	Exits	
North America	−0.285 (0.299)	−0.251 (0.588)	−0.237 (0.909)	−0.213 (0.995)	0.1583 (0.131)
Legality	0.015 (0.794)	0.106 (0.308)	0.069 (0.224)	0.234** (0.012)	0.096* (0.072)
Market Liquidity	0.347*** (0.010)	0.691*** (0.000)	0.207 (0.436)	0.569** (0.048)	0.547*** (0.001)
MTB	−0.05 (0.849)	1.281*** (0.000)	−0.114 (0.743)	1.403*** (0.000)	0.226 (0.585)
Deal Activity	−0.075 (0.686)	0.007 (0.949)	−0.358 (0.132)	0.052 (0.797)	−0.331* (0.075)
GDP per capita	−0.022 (0.880)	−0.183* (0.093)	−0.218 (0.136)	−0.658** (0.000)	0.193 (0.093)
North America × market liquidity	−	−	−0.0379 (0.293)	−0.185** (0.022)	0.411** (0.033)
North America × deal activities	−	−	0.467 (0.016)	0.284 (0.514)	0.604** (0.018)
North America × MTB	−	−	−0.0594 (0.354)	−0.084* (0.091)	−0.2581 (0.265)
North America × GDP per capita	−	−	0.116 (0.119)	0.309** (0.041)	0.038*** (0.001)
VC Age	0.607** (0.025)	0.036 (0.791)	0.418 (0.141)	0.043 (0.758)	0.335 (0.210)
Old VC	−0.292* (0.054)	−0.171* (0.062)	−0.214 (0.168)	−0.181* (0.054)	−0.226 (0.188)
VC Size	0.017** (0.047)	0.002 (0.744)	0.019** (0.029)	0.001 (0.928)	0.016*** (0.000)
Fund Age	0.102 (0.734)	−0.706*** (0.004)	−0.033 (0.916)	−0.624** (0.012)	0.664** (0.045)
Old Fund	−0.244 (0.229)	0.196 (0.250)	−0.234 (0.248)	0.147 (0.397)	−0.829*** (0.002)
Size Inv	0.107 (0.305)	0.019 (0.847)	0.073 (0.481)	0.004 (0.966)	0.054 (0.476)
TOP25% Size Inv	0.029 (0.379)	−0.02 (0.455)	0.031 (0.371)	−0.022 (0.419)	0.067** (0.031)
Synd	−0.178** (0.033)	0.012 (0.623)	−0.193** (0.022)	0.011 (0.667)	−0.425*** (0.003)
Old Synd	0.177** (0.037)	−0.015 (0.560)	0.181** (0.033)	−0.014 (0.574)	0.365*** (0.007)
Company Age	0.265** (0.018)	0.106 (0.295)	0.269** (0.018)	0.102 (0.321)	0.089 (0.431)
Company Size	−0.298 (0.289)	−0.047 (0.783)	−0.259 (0.366)	−0.021 (0.902)	0.557*** (0.002)
Constant	−1.868** (0.027)	−1.126*** (0.000)	−1.042* (0.083)	−1.576*** (0.000)	
Obs	159	175	159	175	334

and A4 in the Appendix). We focused our discussion above on M&As and IPOs because they represent successful exits and are far more numerous than the less successful MBO exits and liquidations. For MBOs and liquidations, we find that cross-border investments are exited significantly more quickly than domestic investments in Model A IV (corresponding to Model I in Table 7a for M&As and IPOs). The statistically significant negative North American indicator in Model A V suggests that MBO and liquidation exits occur more quickly from North American-based portfolio firms than from those based elsewhere, while there are no differences between any other regions (including the UK & Ireland). Once we include market, legal and economic variables in Models A VII and A VIII, the differences between North America and the other regions are no longer significant. This suggests that the speedier MBOs and liquidation exits from North American investments are due to the distinctive economic, legal and capital market characteristics of the region.

4.3. Effects of other variables

In examining the impact of cross-border and regional effects alongside indicators of market, legal and economic development,

we control for a range of characteristics of the VC firm, the VC deal and the portfolio company. Several of these control variables are statistically significant in all four models presented in Tables 7a and 7b. The age of the VC firm turns out to be insignificant for M&A and only significant at the 10 percent level for IPO exits in some models. By contrast, we find a statistically and economically significant exit-accelerating effect of the variable *Old VC* capturing the age of the oldest quartile of VC firms (and zero otherwise). The negative coefficient of *OLD VC* suggests that the most experienced VC firms are able to reduce the time to exit. Specifically, an additional year of age (experience) among the most experienced firms reduces the time to exit by between 5% and 6% and between 10% and 13%, respectively, for M&A and IPO exits. For MBO and liquidation exits, the effect of the most experienced VC firms (*Old VC*) is in the same direction (accelerating exit) and even larger than for M&A and IPO exits (see Tables A3 and A4). A one-year increase in the age of the oldest VC firms speeds up MBO exits by between 25 and 27 percent (based on the significant coefficients), and liquidation exits by between 16 and 19 percent. This result is similar to the finding of Giot and Schwienbacher (2007) that VC age accelerates liquidation.

The size of the VC syndicate reduces the time to exit for both M&A and IPO exits. For M&A exits, an additional syndicate member reduces the time to exit by between 2% and 3%. This is almost exactly the same as the finding of Giot and Schwienbacher (2007) for US VC deals. For IPO exits, Giot and Schwienbacher report an even larger syndicate effect of 6 percent, double the effect on M&As. By contrast, we find a much smaller, though statistically significant, effect of between 1 and 2 percent for IPOs.

The largest effect (among the control variables) is that of the size of the (logarithm of the) VC investment in the portfolio company, *Size Inv*. A one-unit increase in this variable (equivalent to a £2.718 million increase in the amount invested) reduces the time to IPO exit by between 14 and 17 percent in all four models of Tables 7a and 7b. For the largest investments (as indicated by *TOP25% Size Inv*), this effect is reduced by about 4 to 5 percentage points to between 11 and 13 percent. The impact of *Size Inv* on M&A exits is significant (in Models I and II, i.e. in without controlling for the macro factors) but smaller at between 4 and 5 percent. Giot and Schwienbacher similarly report a significant effect of the investment amount for US IPO exits but of a much smaller magnitude (around 2 percent), and none for trade sales.

In Models III–V, the size of the VC firm has a statistically significant but very small effect on M&A exits (but not on IPOs), reducing the time to M&A by about half a percent for a one-unit increase in the logarithm of the size of the VC firm. The (logarithm of the) size of the portfolio company, by contrast, has a much larger and significant effect on M&A exits (but none on IPOs), delaying the time to exit by between 8 and 10 percent. This suggests that larger portfolio companies take longer to be marketed through a trade sale.

4.4. Extensions and robustness tests

As an extension of our analysis, we estimate a competing risk model that examines the effects of our explanatory variables on the choice between various competing exit routes. The results are shown in Table A2 (Model A III) and Table A3 (Model A VI) in the Appendix. As the competing risk model estimates the relative hazards of competing exit routes (rather than the time to exit), the estimated coefficients need to be interpreted such that a positive sign indicates a greater hazard, and hence quicker exit, relative to the competing option. Bearing in mind this difference, the positive North American coefficient reported in the competing risk Model A III of Table A2 suggests that IPO exits happen particularly quickly compared to M&A exits in North America. This is in line with the cumulative exit rates in Table 6, which show that investments are exited through IPOs more quickly than through M&As in both North America and in all other regions. However, the difference in times to exit between the two routes is more pronounced in North America. The same interpretation applies to the positive North American coefficient in the competing risk model of MBOs relative to liquidation exits shown in Table A3, Model A VI; i.e., compared to liquidations, MBOs (including compulsory repurchases) happen particularly quickly in the case of North American portfolio companies.

Next, we test the robustness of the results of our main analysis in Tables 7a and 7b using alternative measures for stock market liquidity and the nature of the legal system. Our alternative measure of market liquidity is defined as the total value of shares traded divided by market capitalization (rather than divided by gross domestic product). Our results remain qualitatively unchanged. As a further robustness test, we replace the legality index with an alternative measure: an indicator of the type of legal system in terms of civil versus common law (e.g., La Porta et al., 1998; Cumming et al., 2010). We find the time to IPO exit in common law countries is (statistically significantly) shorter than in

civil law countries. However, when we add in the legality index, the civil/common law measure becomes insignificant.¹⁰

Further, we examine alternative measures of VC age, fund age, investment size, market liquidity and syndicate size. In the case of syndication, we use the oldest VCs age instead of the average age of the syndicate members. Similarly, for fund age, we use the age of the oldest fund rather than the syndicate average of fund age. We also measure the investment size for syndicates as the largest syndicate investment instead of the syndicate average. Our results are robust to using these alternative measures. Finally, for companies with multiple financing rounds, our original sample excludes all but the first financing round to avoid giving undue weight to these sample companies. As a robustness check, we re-estimate our models using just the last financing round, and we find our results to be qualitatively unchanged.

5. Conclusion

Our study is motivated by the growing prevalence and importance of cross-border VC (which characterizes VC industries outside the US). We investigate the time taken by VC backers to exit domestic and cross-border investments, focusing on VC firms based in the UK, as more than half the funds of UK VC firms are invested outside their home region (UK & Ireland).

For a sample of 4502 VC exits during 1990–2010, we study the time to exit through alternative routes (IPO, M&A, MBO and liquidation). Our multivariate analysis finds a significant and pronounced difference between cross-border and domestic investments in the time to exit for both M&A and IPO. On closer examination, we find that this difference between domestic and cross-border investments is in fact driven by investments in North America. Our results show that while there is no significant difference between the domestic region (UK & Ireland) and most cross-border regions, the average times to exit more than two thirds shorter in North America than anywhere else, including the domestic region (UK & Ireland).

We explore the impact of macro factors (in terms of the nature and development of the regional economies, capital markets and legal systems). Due to our initial finding of a significant difference between the investments in North America and those elsewhere, we control for the impact of North American investments in order to accurately assess the impact of macro variables on exit times worldwide. Previous cross-country venture-capital studies also note the large proportions of North American observations in cross-country samples and, like us, control for their impact in the analysis (e.g., Cumming et al., 2010; Wang and Wang, 2012).

We find that market liquidity and legality speed up the time to IPO exit, consistent with our predictions and the findings of Black and Gilson (1998), Giot and Schwienbacher (2007) and Schwienbacher (2008) for liquidity, and Cumming et al. (2006) for legality. M&A exits are speeded up by higher stock market valuations in line with our expectations and the findings of Cumming and Johan (2010). Gross domestic product (*GDP per capita*) slows down the time to exit for both M&A and IPO consistent with the findings of Wang and Wang (2012) and the catch-up ('convergence') effect that predicts that poorer regions outperform richer regions. However, the North American indicator remains significant. We address this by further controlling for the distinctiveness of the North American subsample and by exploring whether the impact of the macro factors differ between North America and the other regions. After including interaction terms of each of the four macro variables with the North America indicator, we find that liquidity and some other macro factors indeed affect the time

¹⁰ Detailed results are not reported but are available from the authors.

to exit differently in North America. By failing to control for this differential impact we underestimate the true accelerating effect of *Market Liquidity* on IPO exit across all the other regions outside North America.

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Appendix A.

References

- Aizenman, J., Kendall, J., 2012. The internationalization of venture capital. *Journal of Economic Studies* 39, 488–511.
- Berkowitz, D., Pistor, K., Richard, J., 2003. Economic development, legality and the transplant effect. *European Economic Review* 47, 165–195.
- Black, S., Gilson, J., 1998. Venture capital and the structure of capital markets: banks versus stock markets. *Journal of Financial Economics* 47, 243–277.
- Brander, J., Amit, R., Antweiler, W., 2002. Venture-capital syndication: improved venture selection vs. the value added hypothesis. *Journal of Economics and Management Strategy* 11, 423–452.
- British Venture Capital Association, 2010. Report on Investment Activity, PricewaterhouseCoopers.
- Cumming, D., 2008. Contracts and exits in venture capital finance. *The Review of Financial Studies* 21, 1947–1982.
- Cumming, D., Dai, N., 2010. Local bias in venture capital investments. *Journal of Empirical Finance* 17, 362–380.
- Cumming, D., Johan, S., 2008. Pre-planned exit strategies in venture capital. *European Economic Review* 52, 1209–1241.
- Cumming, D., Johan, S., 2010. Venture capital investment duration. *Journal of Small Business Management* 48, 228–258.
- Cumming, D., MacIntosh, J., 2003. Venture capital exits in Canada and the United States. *University of Toronto Law Journal* 53, 101–120.
- Cumming, D., Fleming, G., Schwienbacher, A., 2006. Legality and venture capital exits. *Journal of Corporate Finance* 12, 214–245.
- Cumming, D., Schmidt, D., Walz, U., 2010. Legality and venture capital governance around the world. *Journal of Business Venturing* 25, 54–72.
- Demirgüç-Kunt, A., Levine, R., 1996. Stock markets, corporate finance, and economic growth: an overview. *World Bank Economic Review* 10, 223–239.
- Devigne, D., Vanacker, T., Manigart, S., Paeleman, I., 2013. The role of domestic and cross-border venture capital investors in the growth of portfolio companies. *Small Business Economics* 40, 553–573.
- Dittmar, A., Dittmar, R., 2008. The timing of financing decisions: an examination of the correlation in financing waves. *Journal of Financial Economics* 90, 59–83.
- Espenlaub, S., Khurshed, A., Mohamed, A., 2012. IPO survival in a reputational market. *Journal of Business Finance and Accounting* 39, 427–463.
- Espenlaub, S., Khurshed, A., Mohamed, A., 2014. Does cross-border syndication affect venture capital risk and return? *International Review of Financial Analysis* 31, 13–24.
- European Commission, 2013. Report of Expert Group on Removing Tax Obstacles to Cross-border Venture Capital Investments.
- Fine, P., Gray, R., 1999. Proportional hazard model for the sub distribution of a competing risk. *Journal of the American Statistical Association* 94, 496–509.
- Giot, P., Schwienbacher, A., 2007. IPOs, trade sales and liquidation: modelling venture capital exits using survival analysis. *Journal of Banking & Finance* 31, 697–702.
- Gompers, P., 1996. Grandstanding in the venture capital industry. *Journal of Financial Economics* 42, 133–156.
- Gompers, P., Lerner, J., 2001. The venture capital revolution. *Journal of Economic Perspectives* 2, 145–168.
- Hensler, D., Rutherford, R., Springer, T., 1997. The survival of initial public offerings in the aftermarket. *Journal of Financial Research* 20, 93–110.
- Jain, B., Kini, O., 2000. Does the presence of venture capitalists improve the survival profile of IPO firms? *Journal of Business Finance and Accounting* 27, 1139–1176.
- Jelic, R., Wright, M., 2011. Exits, performance, and late stage private equity: the case of UK management buyouts. *European Financial Management* 17, 560–593.
- Kandel, E., Leshchinskii, D., Yuklea, H., 2011. VC Funds: Aging Brings Myopia. *Journal of Financial and Quantitative Analysis* 46, 431–457.
- Kleinbaum, D., 1996. *Survival Analysis: A Self-learning Text*. Springer Verlag, New York.
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R., 1998. Law and Finance. *Journal of Political Economy* 106, 1113–1155.
- Lerner, J., 1995. Venture capitalists and the decision to go public. *Journal of Financial Economics* 35, 293–316.
- Mäkelä, M., Maula, M., 2005. Cross-Border venture capital and new venture internationalization: An isomorphism perspective. *Venture Capital: An International Journal of Entrepreneurial Finance* 7, 227–257.
- Mäkelä, M., Maula, M., 2006. Interorganizational commitment in syndicated cross-border venture capital investments. *Entrepreneurship Theory and Practice* 30, 273–298.
- Meggison, W., Weiss, K., 1991. Venture capital certification in initial public offerings. *Journal of Finance* 46, 879–893.
- Pearce, R., Barnes, S., 2006. *Raising Venture Capital*. Wiley Finance, UK.
- Rhodes-Kropf, M., Robinson, D., Viswanathan, S., 2005. Valuation waves and merger activity: the empirical evidence. *Journal of Financial Economics* 77, 561–603.
- Schertler, A., Tykvová, T., 2011. Cross-border venture capital flows and local ties: evidence from developed countries. *The Quarterly Review of Economics and Finance* 51, 36–48.
- Schwiebacher, A., 2008. Venture capital investment practices in Europe and the United States. *Financial Markets and Portfolio Management* 22, 195–217.
- Shleifer, A., Vishny, R., 2003. Stock market driven acquisitions. *Journal of Financial Economics* 70, 295–311.
- Shumway, T., 2001. Forecasting bankruptcy more accurately: a simple hazard model. *Journal of Business* 74, 101–124.
- Wang, L., Wang, S., 2012. Economic freedom and cross-border venture capital performance. *Journal of Empirical Finance* 19, 26–50.